

Quick Pharma VSM

Learn how to use Pharma VSM application for plant level value stream mapping of pharmaceutical production that has upstream fermentation and downstream extraction phases.



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How to Use this File

This file contains the reading materials and the exercise pages from the course (title on previous page). While the course can only be taken on a computer, this booklet can be useful for note taking and later for refresher training.

This booklet is designed for on screen and print use. For on screen use, we recommend Acrobat Reader with the page display set to "Single Page View". If you are using this booklet on-screen while going through the exercises in eVSM, a second monitor is very helpful.

For hardcopy use, print the file on 8.5x11 or A4, and bind along the long edge.

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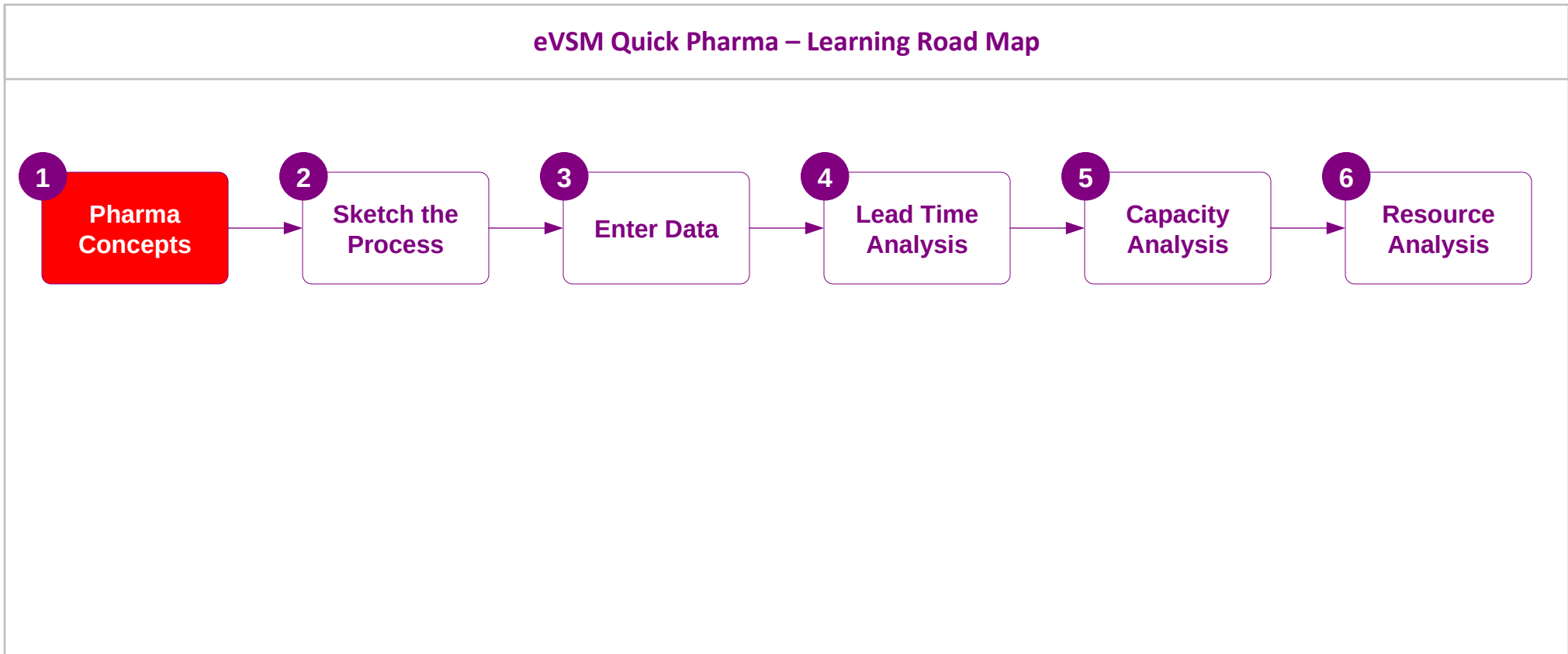
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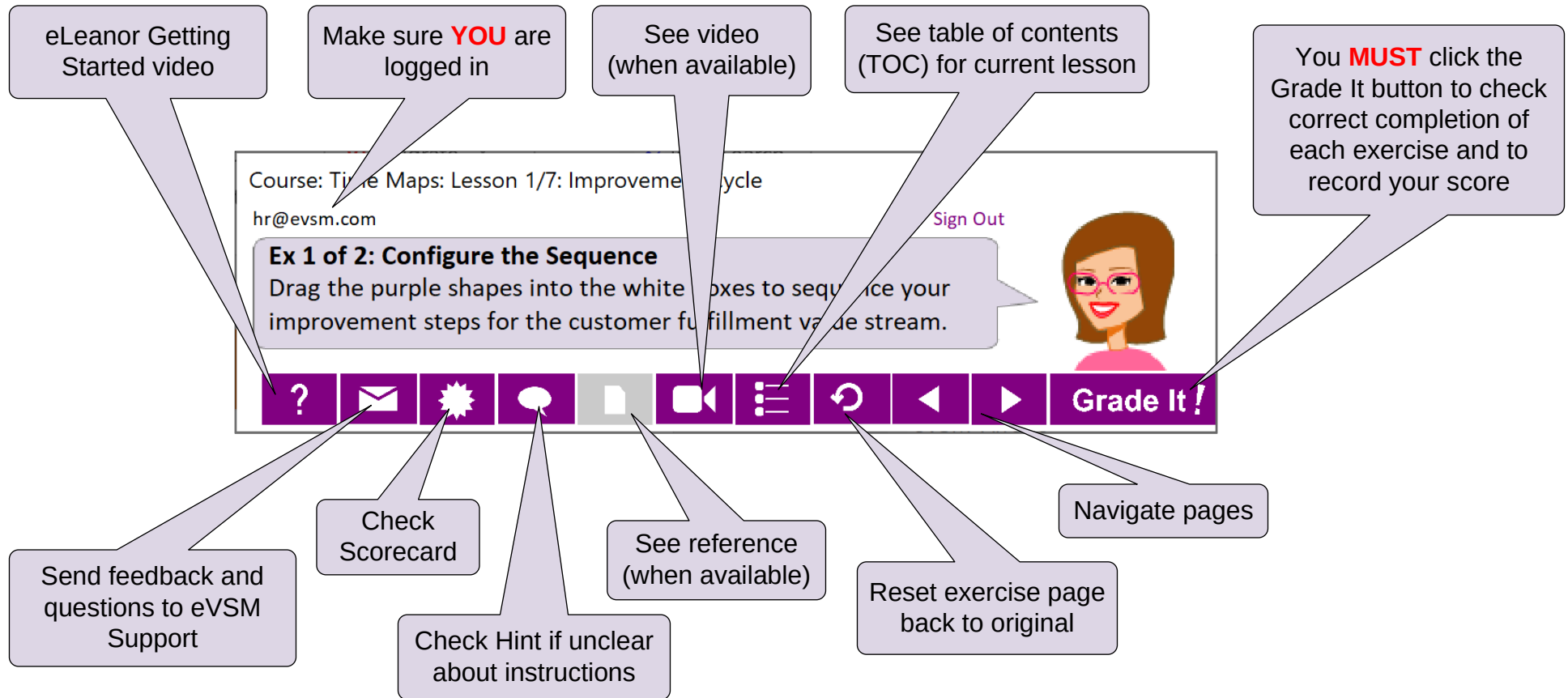
Introduction to Pharmaceutical Production Concepts

The Quick Pharma application is used for plant level value stream mapping of pharmaceutical API production. The value stream starts with a vial of cell culture and ends with the active pharmaceutical ingredient (API). The application standardizes the variables, shapes, equations and charts used in such maps.

This first lesson covers some of the base concepts for Pharma value stream mapping.



Working with the eLearner Control Panel



Important Notes

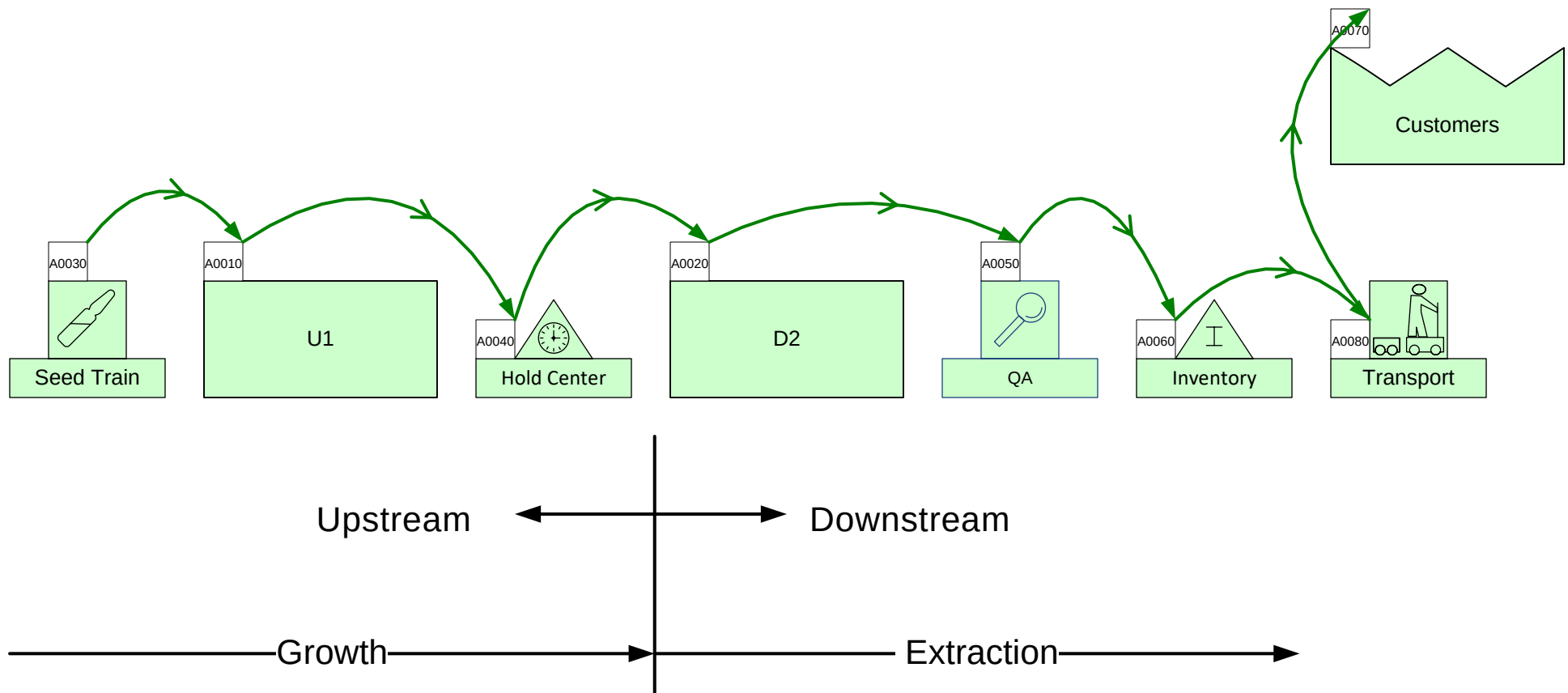
1. Make sure you have a good eLearner environment: large screen PC, 1280x720 resolution minimum, physical mouse with scroll wheel.
2. A PDF copy of the course is included in the downloaded fileset. It is very helpful to view this on a second while going through the training.
3. When you complete an exercise, you **MUST** click the "Grade It" button.
4. You WILL lose points if you get an exercise wrong the first time.
5. If you are stuck on an exercise, check the Hint. If that does not help, go back and review the preceding Readme pages. If you are still unsure, click the Feedback button in the eLearner panel and ask your question.

Basic Pharma API Production Map

Let's use a simple map to discuss some of the Pharma API (Active Pharmaceutical Ingredient) production concepts.

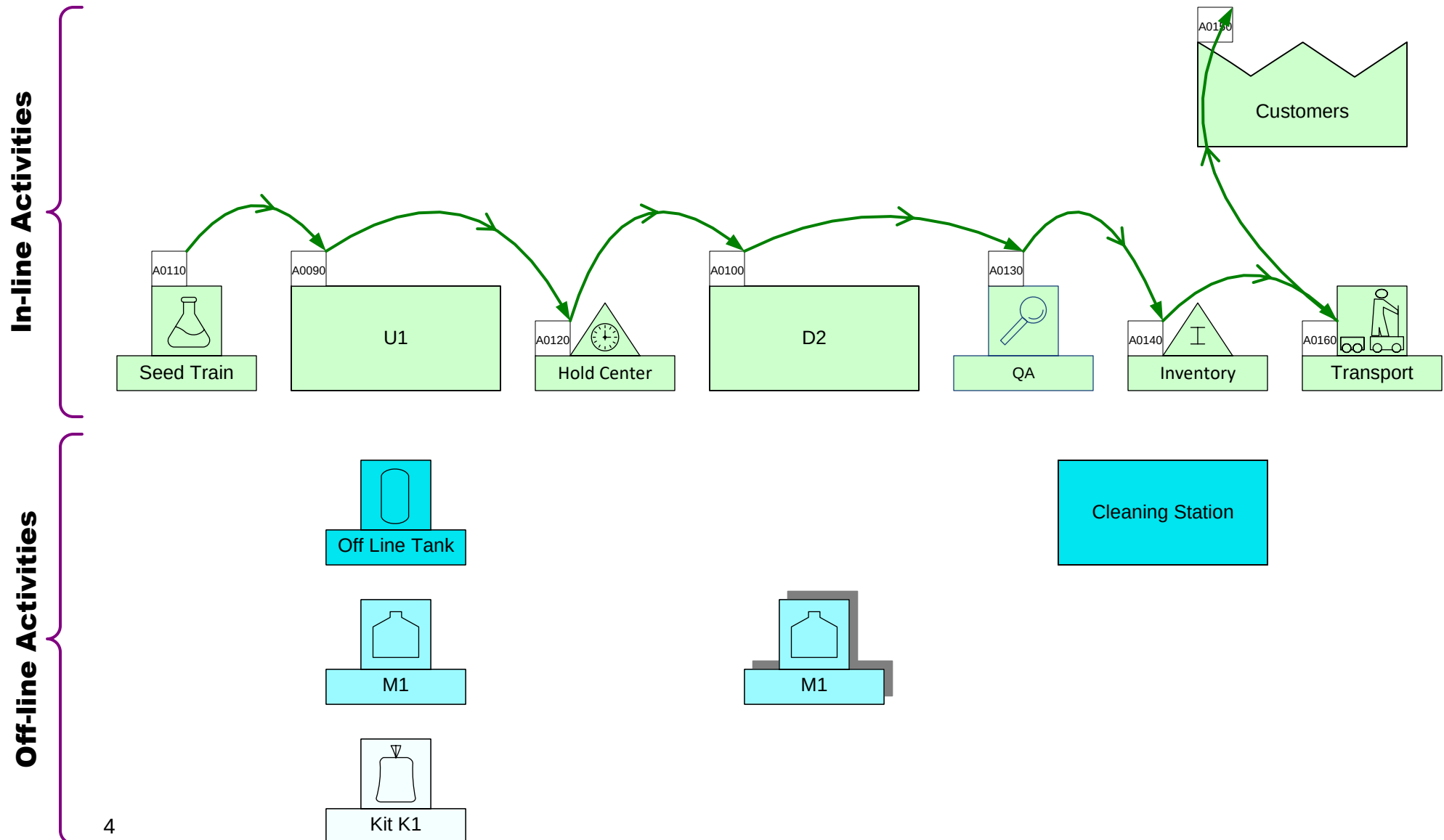
Let's look at a Bio API where Production starts with an ampoule containing the cells and the growth of these in the "Seed Train" using flasks, wave bags, and tanks as the volume increases. This is all part of the upstream operation and can be mapped with one or more seed train shapes and one or more in-line activity shapes like U1, below.

The resulting materials after the growth phase is put in a holding tank from which batches are withdrawn as needed for the downstream extraction phase. Each batch created down stream is termed a "run".



Basic Pharma API Production Map (Contd.)

The in-Line activities track the main pharma product from cell culture through to finished goods and include cell growth, extraction and in-line hold and quality checks. The In-Line activities impact the overall lead time from cell culture to released product. The off-line activities like media creation, storage in tanks are important but are assumed to not delay the in-line activities or impact lead time. The off-line activities are just placed visually where they directly support the in-line activities (like the kit, media, tank below) or off to the side where they are supporting the line in general (like the cleaning station below)



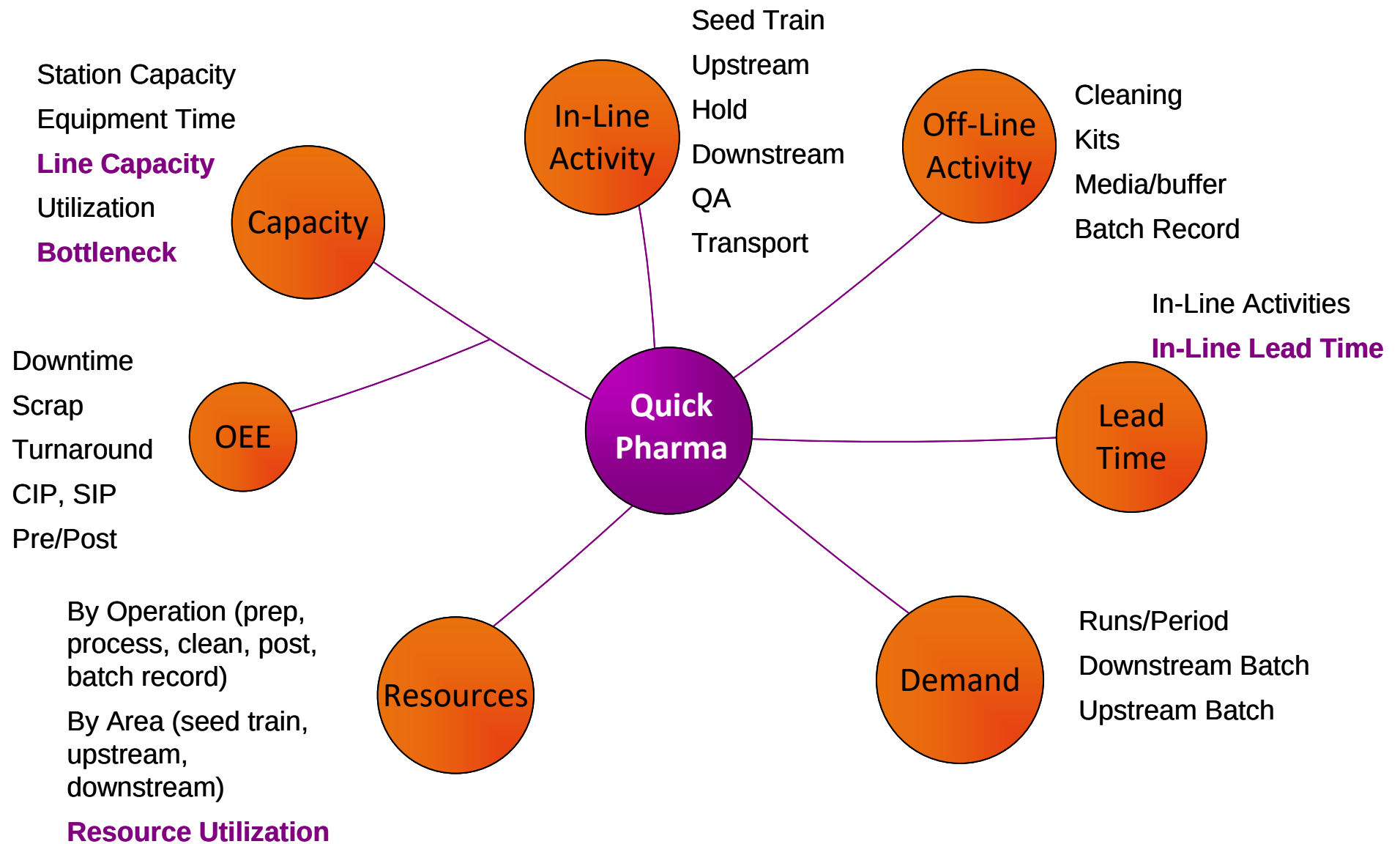
Which of the following is true for "Off-Line" activities?

Select all 3 that are true.

- ☐ They support the in-line activities
- ☐ They are assumed to be managed so they don't slow the in-line activities
- ☐ They consume resource
- ☐ They contribute to the in-line lead time calculation

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Quick Pharma Concepts

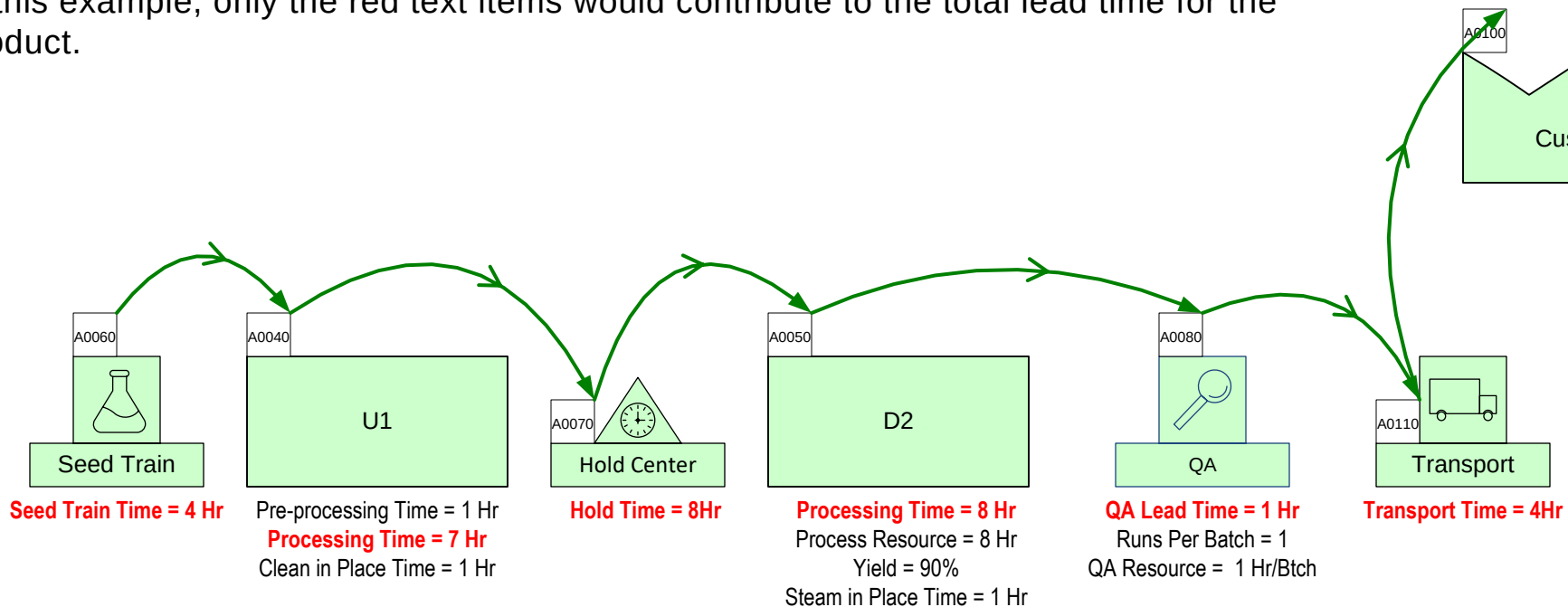


Lead Time

Lead time is one of the many calculations that Quick Pharma is capable of computing. The calculation is simply the time it takes to create a product from the start to the end of the process.

Lead Time is the time required for the product to go through the process. It does not include equipment time (pre-process setup, or post strip-down, cleaning, etc.) or people resource time.

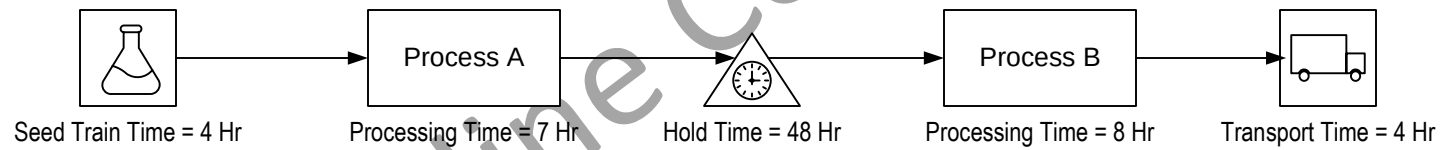
In this example, only the red text items would contribute to the total lead time for the product.



$$\text{Lead time} = 4 + 7 + 8 + 8 + 1 + 4 = 32 \text{ Hr}$$

What is the Lead Time for the Process below?

- ☐ 50 Hr
- ☐ 25 Hr
- ☐ 71 Hr
- ☐ 15 Hr



Lead Time Components

Let's imagine a batch going through one station (an in-line activity). The associated times are shown via the blocks below:

Pre	Process	Post	CIP	SIP
-----	---------	------	-----	-----

The station is prepped to receive the batch (Pre step), then it receives and processes the batch (Process step), followed by Post time (Post step), Clean In Place (CIP) and Steam In Place (SIP).

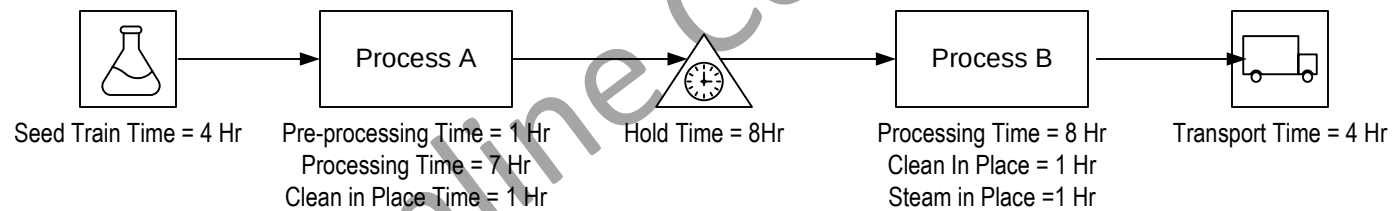
If I imagine a molecule in the batch, it enters the station at the start of the Process step and leaves at the end of the step. So for measuring **lead time** associated with that molecule, we only use the Process time.

In addition to stations that process a batch, other in-line activities that impact the lead time are in-line QA, Transport, Holds and Wait times in Inventory:

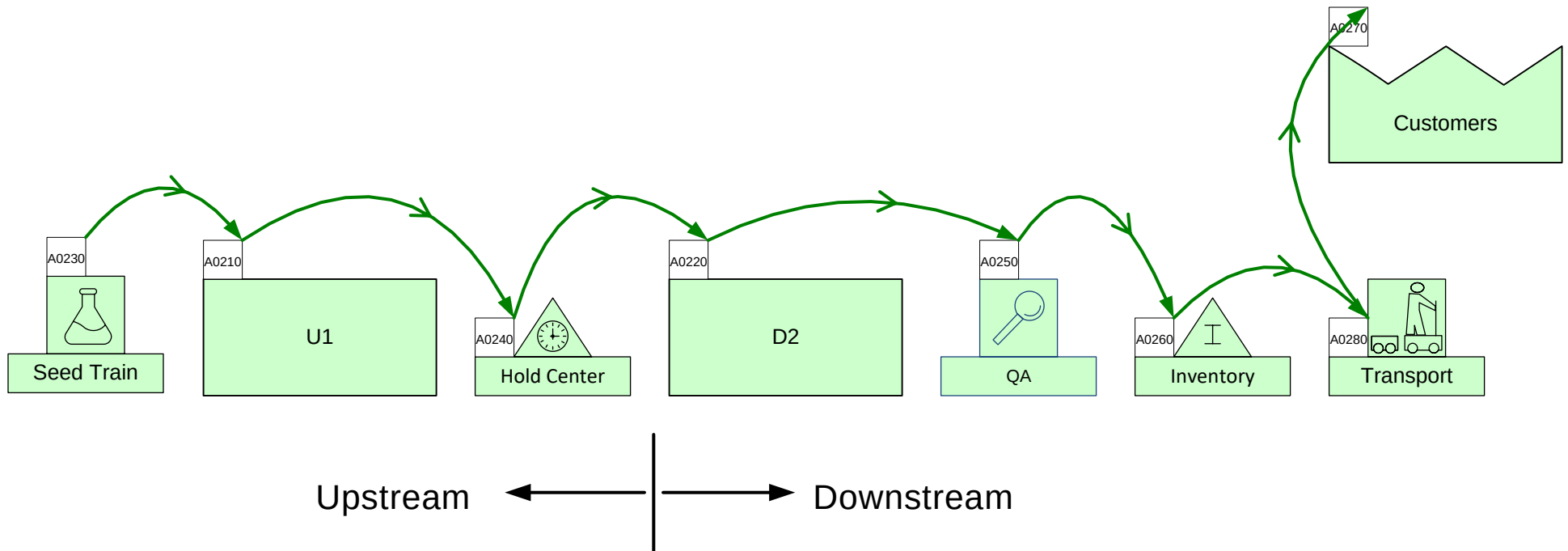
Process	In-Line QA	Hold	Transport	Wait in Inventory
---------	------------	------	-----------	-------------------

What is the Lead Time for this Process?

- ☐ 28 Hr
- ☐ 31 Hr
- ☐ 37 Hr
- ☐ 51 Hr



Runs & Batches



A batch of product at the end of the downstream operation is termed a “Run”.

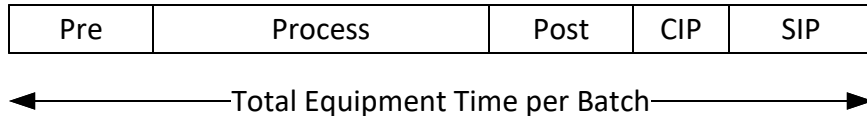
Note that a downstream batch does not always correspond to an upstream batch. The holding tank between upstream and downstream decouples the batch flow.

Similarly off-line media and kits made to support the in-line activities are made in their own batch sizes. Since we think of demand as the number of runs per time period we have to convert between any local batch size and how many runs that batch size will support.

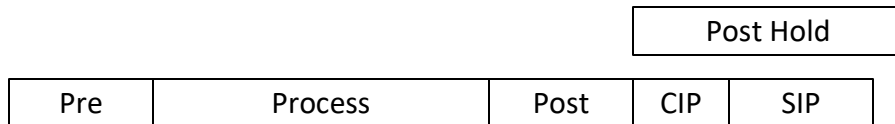
Finally, we have to understand the downstream finished weight per run so we can understand the weight of API produced per period.

Equipment Time

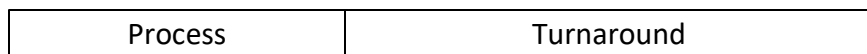
The equipment is tied up from the beginning of the Pre-Step to the end of the SIP step. So for **equipment time** calculation, all the below times are combined together. So this station can process one local batch every combined time interval.



Depending on the station and its usage, some of the time blocks will be unnecessary and there may be some variants. One example is the need for the equipment to be in hold mode for a minimum period after the “Post” step. We call this the “Post Hold” time and it runs in parallel with the CIP and SIP steps. So the equipment time component used is the maximum of Post Hold and CIP+SIP.



Sometimes, its not useful to break out all the steps and instead the concept of Turnaround time is used. This is the time from the end of the Process step until the next Process step.



In this case, the equipment time for one local batch is based on the sum of the process time and the turnaround time.

Capacity

The equipment **Capacity** is based on:

- Equipment time per batch
- Available time of the equipment
- Number of runs per batch

Capacity calculation is simply:

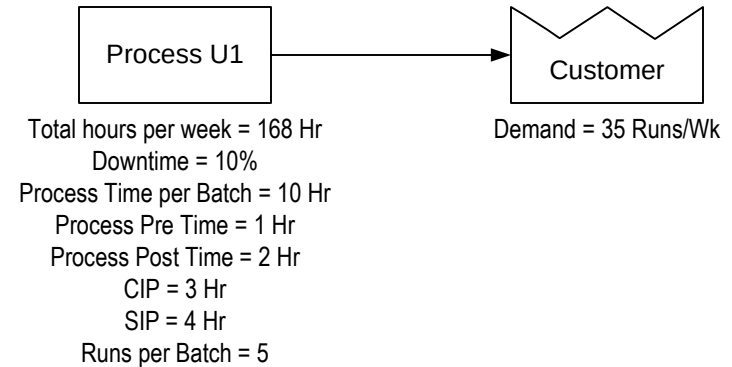
$$\text{Capacity} = \frac{[\text{Available Time}]}{[\text{Time per Run}]}$$

Loss of capacity occurs if the [Available Time] is compromised, eg. Due to downtime, unplanned events (UPes), etc.

Similarly, capacity will be reduced if the [Time per Run] increases. Examples include steam in place (SIP), clean in place (CIP), other pre/post activities, and slower processing.

And, ofcourse capacity is affected by Yield%.

Example Calculation



Capacity Calculation for Process U1

$$\text{Capacity} = \frac{168 * ((100-10)/100)}{(10 + 1 + 2 + 3 + 4) / 5}$$

$$\text{Capacity} = 37.8 \text{ Runs/Wk}$$

**Which of the following times impact the capacity of equipment?
Tick all 4 that are TRUE.**

- ☐ Processing Time
- ☐ Pre Time
- ☐ CIP (Clean in Place)
- ☐ Wait time in inventory
- ☐ Turnaround Time

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Given the data below, what is the Capacity at Process A?

- ☐ 11 Runs/Wk
- ☐ 10 Runs/Wk
- ☐ 14 Runs/Wk
- ☐ 8 Runs/Wk

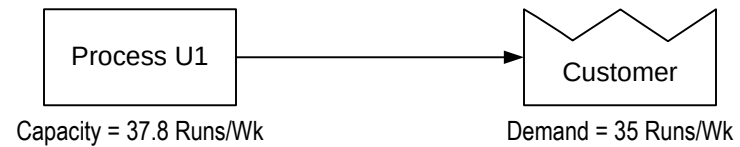
Plant Production Hours = 120 Hr/Wk



Capacity Utilization

Utilization is the percent of capacity being used by the current customer demand.

Example



Utilization Calculation for Process U1

$$\text{Utilization} = \frac{[\text{Demand}]}{[\text{Capacity}]}$$

$$\text{Utilization} = \frac{35}{37.8}$$

$$\text{Utilization} = 93\%$$

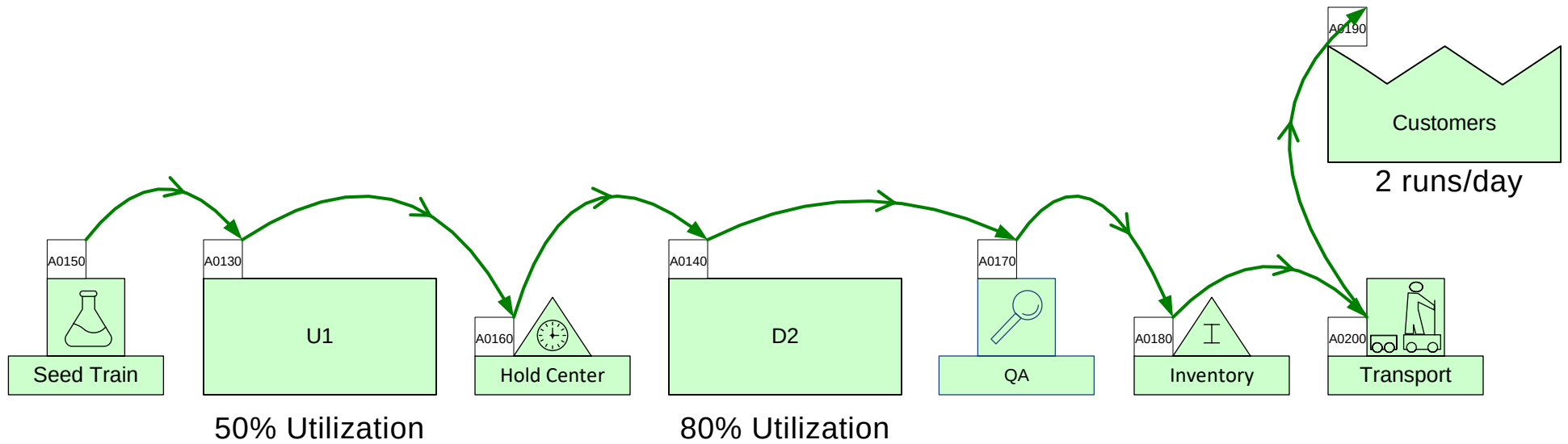
What is the Utilization at Process C?

- ☐ 50%
- ☐ 33%
- ☐ 200%
- ☐ 25%



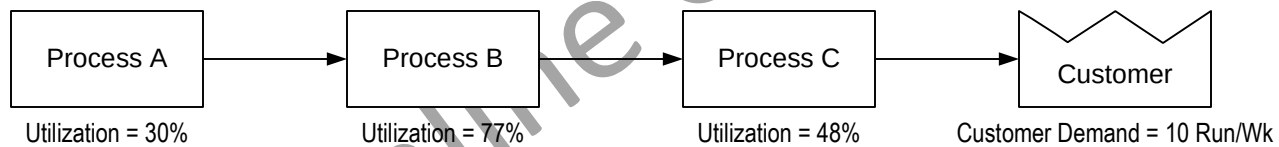
Line Capacity

In the example below, let's say that for a current demand of 2 runs/day, Station U1 had a utilization of 50% and Station D2 had a utilization of 80%. The **line capacity** is assumed to be governed by the highest utilized in-line station and is calculated as current line demand / current max utilization. In our example this would be $2/0.8 = 2.5$ runs/day.



What is the Line Capacity of this Value Stream?

- ☐ 7 Run/Wk
- ☐ 18 Run/Wk
- ☐ 17 Run/Wk
- ☐ 13 Run/Wk



Resource Calculation

Most of the VSM icons have associated resource data and it can be one of two types:

1. Resource per batch
2. Resource per time period

If both values are entered, then the overall resource required is calculated as the sum of the two.

The “Resource per batch” is converted to a resource per period value by multiplying it by the batches going through that activity in a given time period to meet customer demand.

For an in-line activity, there is resource associated with each step in the activity as show below.

Pre	Process	Post	CIP	SIP
2Hr/Batch	2Hr/Batch	0.5Hr/Batch	1Hr/Batch	1Hr/Batch

In this example the total resource required per batch at that activity is 6.5 Hrs/Batch calculated by summing the step values .

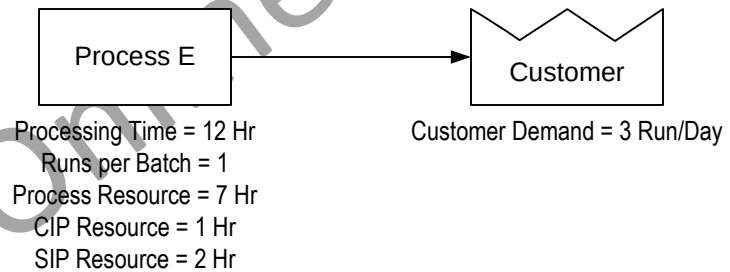
If the number of runs to meet customer demand was 5 per day and each batch was sufficient for 1 run, then the resource required per day would be $6.5 \times 5 = 32.5$ Hr/Day.

Resource Utilization

$$\text{Resource Utilization} = \frac{\text{Resource Required}}{\text{Resource Allocated}}$$

What is the Resource Required at Process E?

- ☐ 10 Hr/Day
- ☐ 22 Hr/Day
- ☐ 66 Hr/day
- ☐ 30 Hr/Day



You learned:

- The key concepts of Pharma value stream mapping including the icons used, upstream/downstream, and in-line vs off-line modeling.
- Pharma value stream analyses for lead time, capacity, and resource.

eVSM Quick Pharma – Learning Road Map

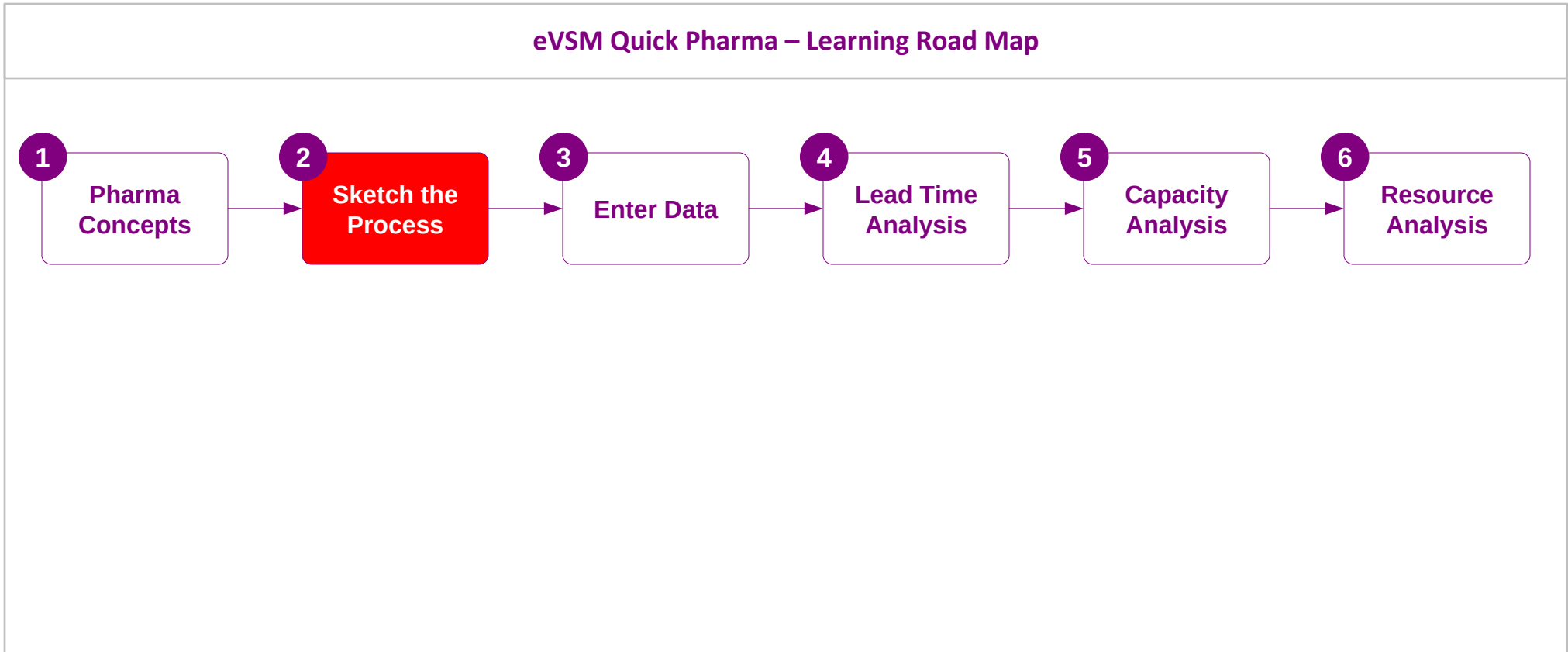


What's next:

In the next several lessons, you will learn how to sketch the material flow for the value stream, how to add data, and then how to set up the automated calculations to help answer what-if questions.

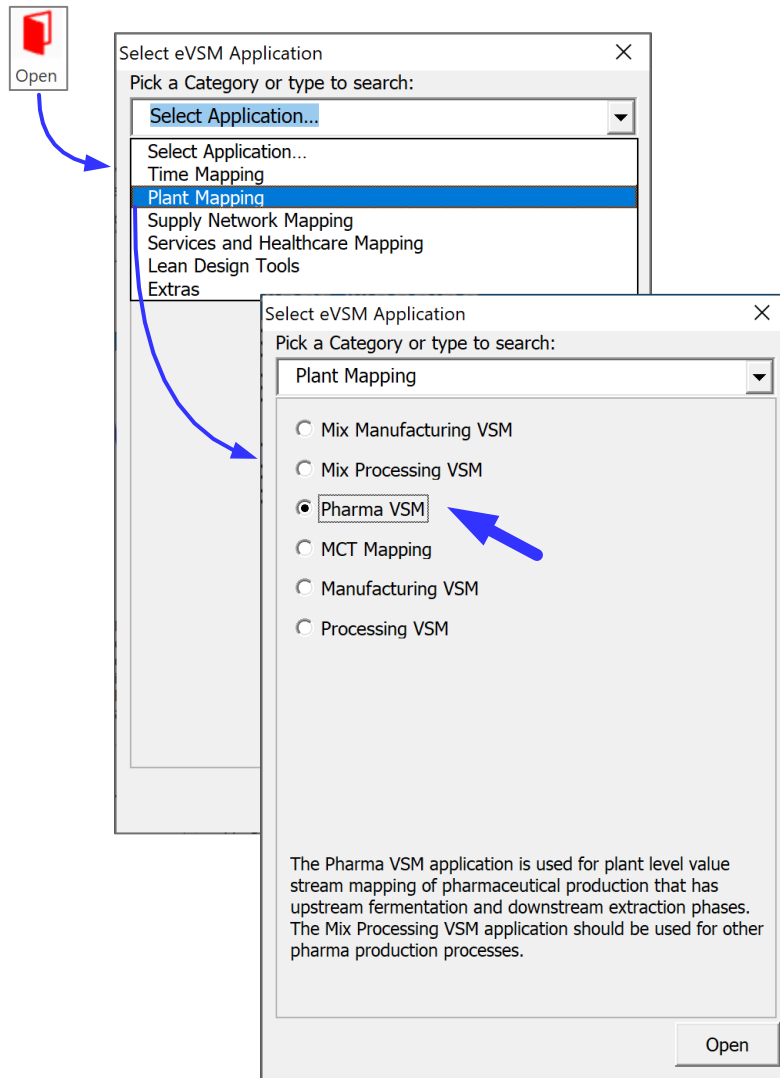
Sketch the Pharma Process Flow

This lesson shows how to access the Quick Pharma stencils and quickly sketch the in-line and off-line material flow for value streams.

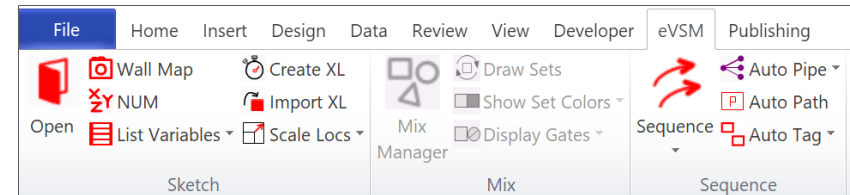


Start eVSM and Open the Quick Pharma Stencils

Opening the Quick Pharma Stencils

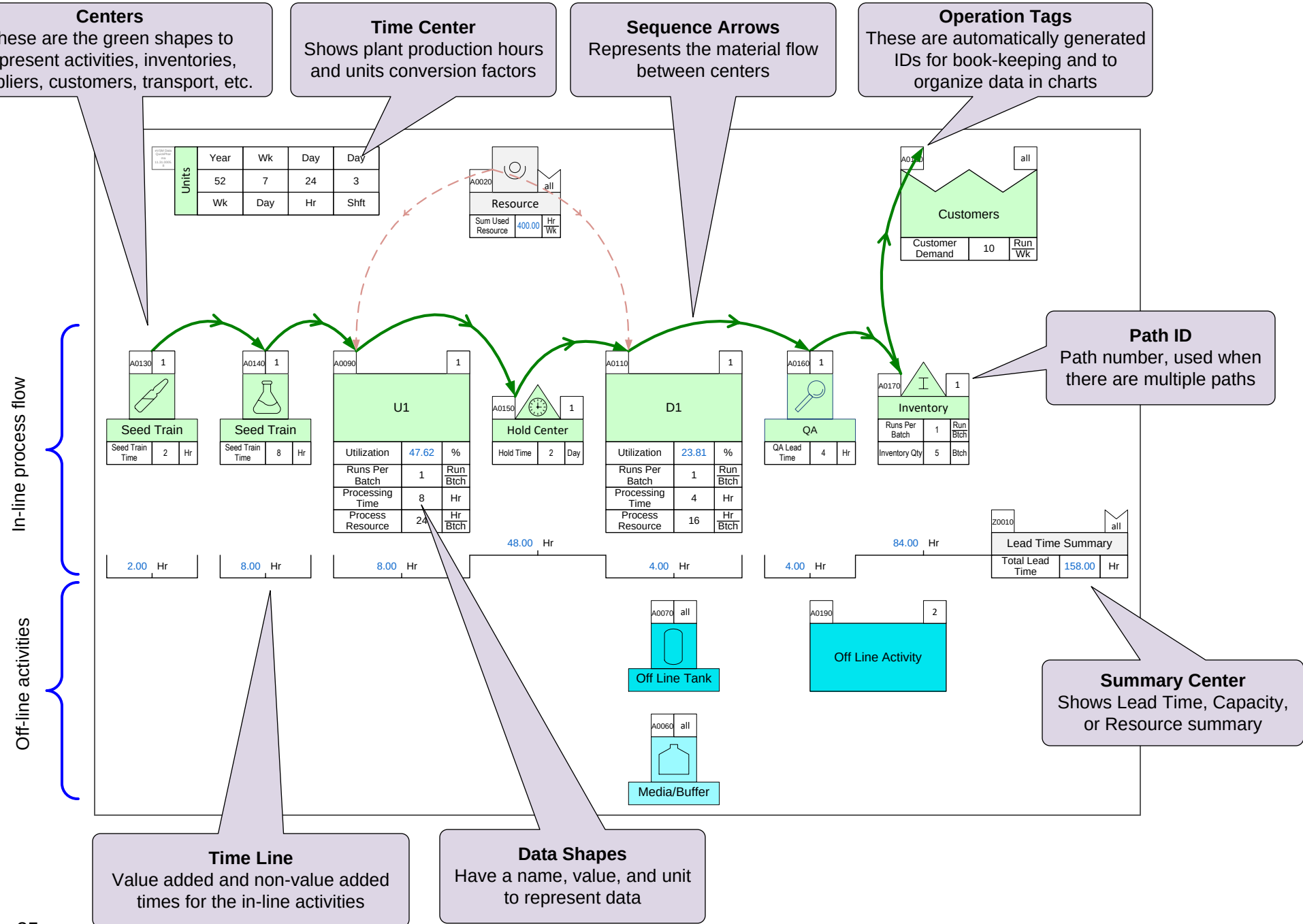


eVSM Mix Toolbar Functions



The Quick Mix toolbar functions are only available when Mix stencils are open. For Quick Pharma, this section will stay greyed out.

Essential Terminology...



eVSM Quick Pharma Application

The Quick Pharma application is delivered as two compatible Visio stencils.

The first is the “Sketch Pharma” stencil that can let you draw the main map using icons but without data shapes. The data shapes can be added later.

The second is the “Quick Pharma” stencil that allows you to draw with the data shapes from the outset. It also has data shape add-ons (yellow icons) to support further analytics.

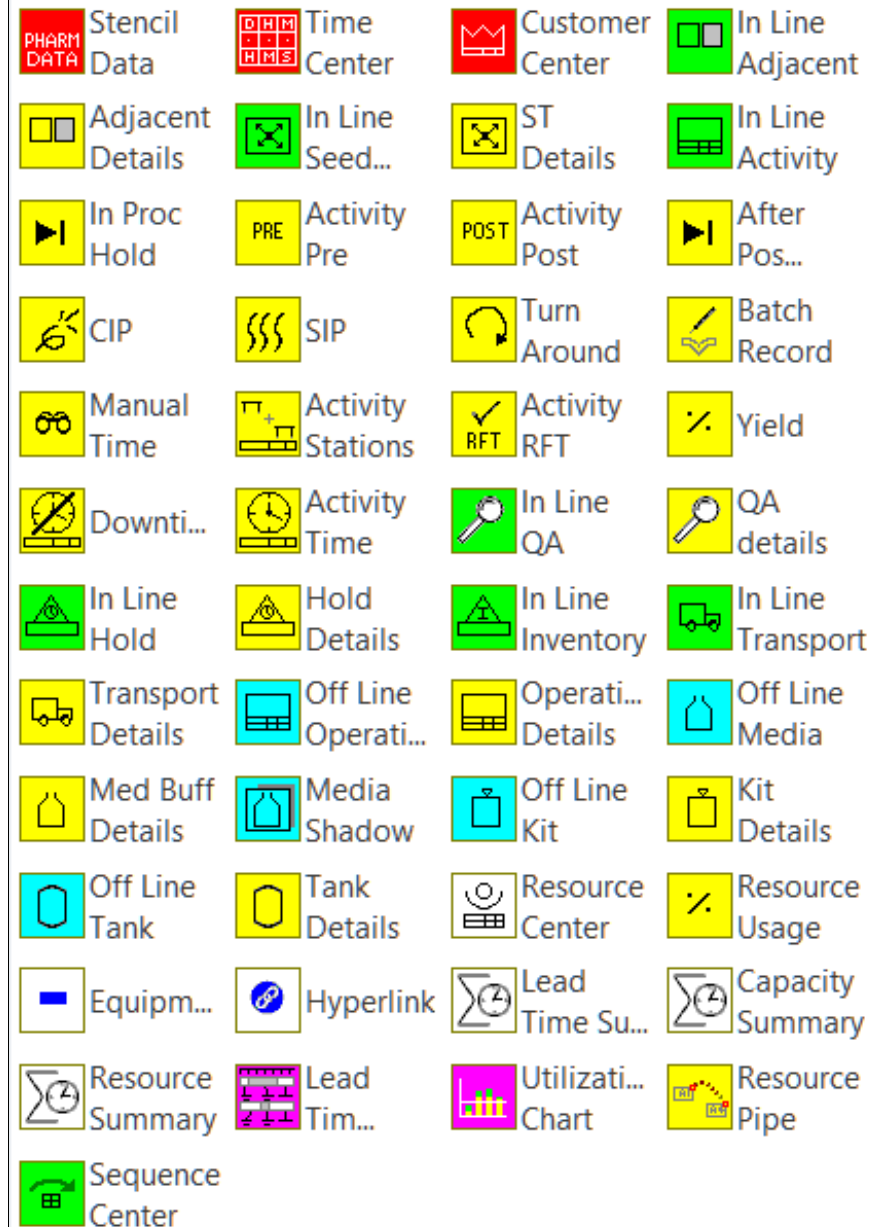
Sketch Pharma

Drop Quick Shapes here



Quick Pharma

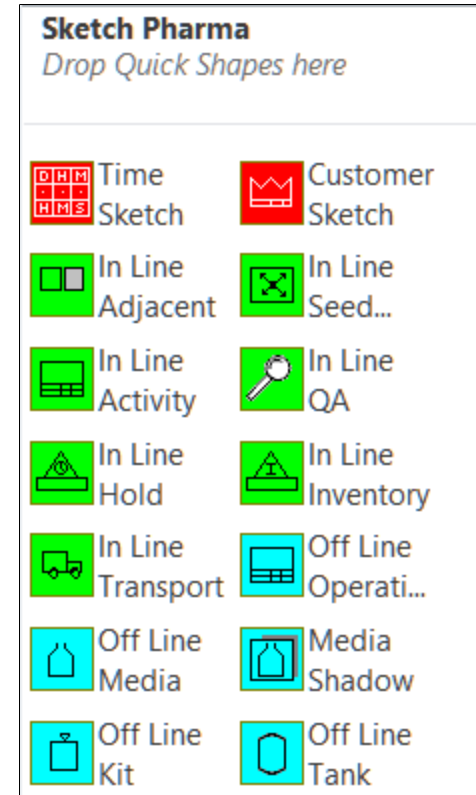
Drop Quick Shapes here



In-Line and Off-Line Operations

In the pharma production process there are operations that impact the overall lead time from the ampoule in the seed train to the finished drug substance after extraction. These are called In-Line activities and are represented as the green icons in the Sketch Pharma or Quick Pharma stencils.

There are also supporting operations like media creation that are essential but are assumed to be timed so that they run in parallel to the in-line activities and are managed such that they do not impact the overall lead time. These are called the Off-Line activities and are represented in blue in the Sketch Pharma and Quick Pharma stencils.

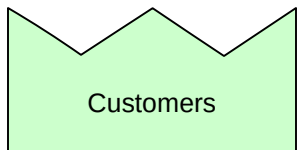


Quick Pharma Value Stream Mapping Icons

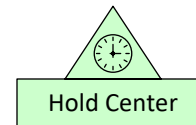
The Bio pharma map can be thought of as a combination on in-line and off-line activities.

The in-Line activities tracks the movement of the main pharma material from cell culture through to product release and includes cell growth, extraction and in-line hold, transport and quality checks. The In-Line activities impact the overall lead time from cell culture to released product. The off-line activities like kit creation, media creation, storage in tanks are assumed to not delay the in-line activities and impact lead time. The in-Line activity icons and the customer center are shown below. The off-line icons are shown on the next page.

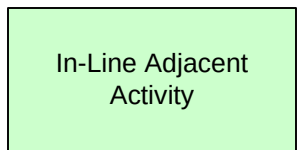
In-Line Icons



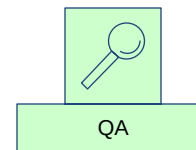
Customer Shape – Used to indicate demand.



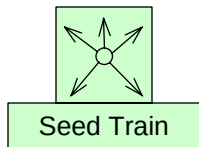
Shape used to represent in-line hold times.



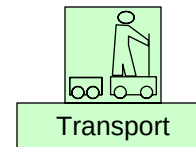
If you choose to split the map (for example upstream and downstream) than this shape is used to summarize the adjacent map so its impact can be seen.



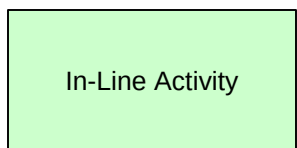
Shape used to represent in-line QA.



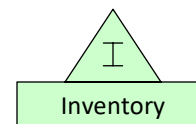
You can use a single shape to represent the whole seed train or multiple ones to represent the different stages from a vial to a tank. Icon variations support graphics like ampoule, wave bag, flask, tank.



Shape used to represent in-line Transport. Icon variations support other transport types.



This shape represents any general upstream or downstream activity and allows for pre, processing, and post steps.

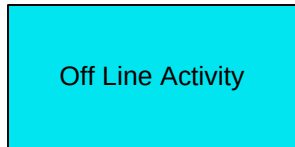


Shape used to represent in-line Inventory storage.

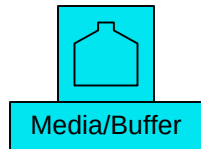
Quick Pharma Value Stream Mapping Icons (Contd.)

Off-line activities support the in-line production but are assumed to be managed such that the in-line production lead time is not impacted.

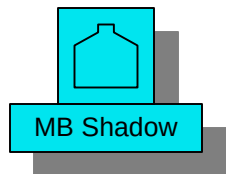
Off-Line Icons



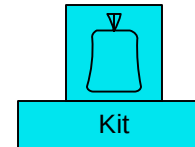
Shape represents a general off-line activity. For example you could have an area dedicated to cleaning small equipment used frequently in production.



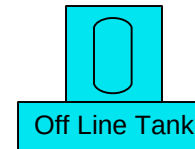
Shape represents media/buffer needed for an in-line activity.



If the same media is used in several places, it can be represented once with details by the media/buffer shape and then everywhere else by the MB (media buffer) shadow shape having the same name.

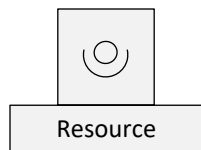


Shape represents kit used to make media/buffer.



Shape represents a storage tank supplying a production step.

Auxiliary Icons



Shape representing a pool of staff. It can be connected to the activities served by the staff and will estimate the minimum staff required.



Equipment ID Tag. For visual purposes only.



Hyperlink Shape. Position on map and then insert a hyperlink to another Visio page or an external file. For navigation purposes only.

Drag out the sketch icons shown below and overlay them on the grey images

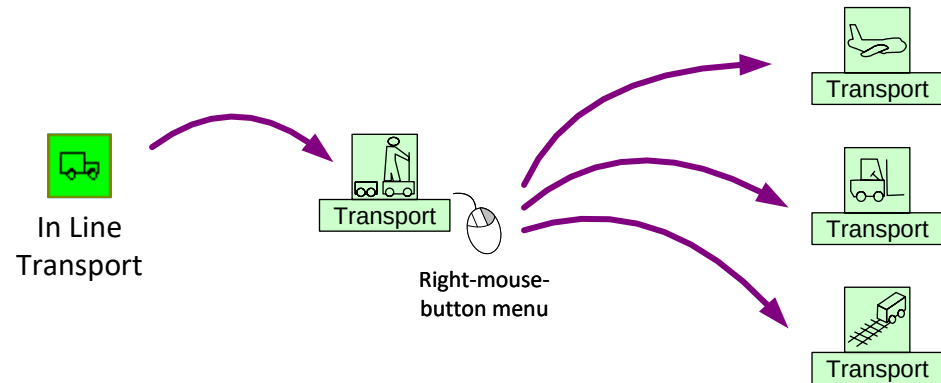


Refresher Tips

Topics below were covered in more detail in the Time Mapping course. The notes here serve as a refresher.

1. Icon Stacks

Icons such as Seed Train, Activity, and Transport belong to families of similar icons. You can switch to other shapes within the same family (e.g. Forklift to Truck) with the “Change Shapes” command in the right-mouse menu of the icon.



2. Page Resizing

To re-size the Visio drawing page, hold the Ctrl key and drag any page edge to the new size.

3. Adding Sequence Arrows

Sequence arrows are used to show the material/information flow between activities.

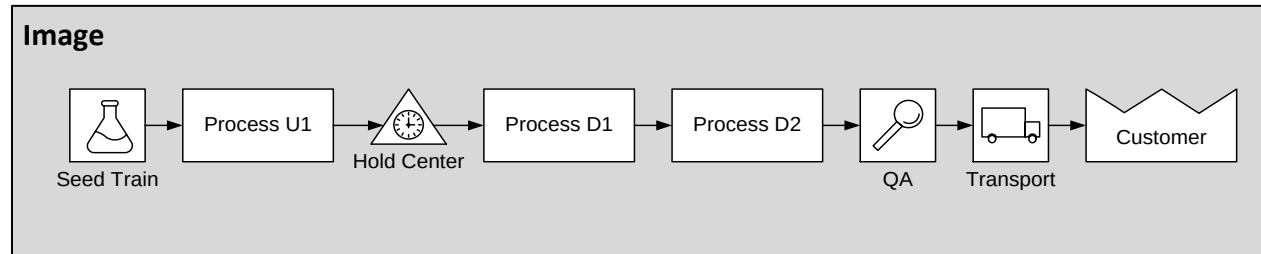
Sequence arrows are mandatory. Without these, the eVSM automated calculations cannot work.

To create and connect Sequence arrows, hold down the Ctrl key, select the centers in the required sequence, and then click the Sequence button in the eVSM ribbon.



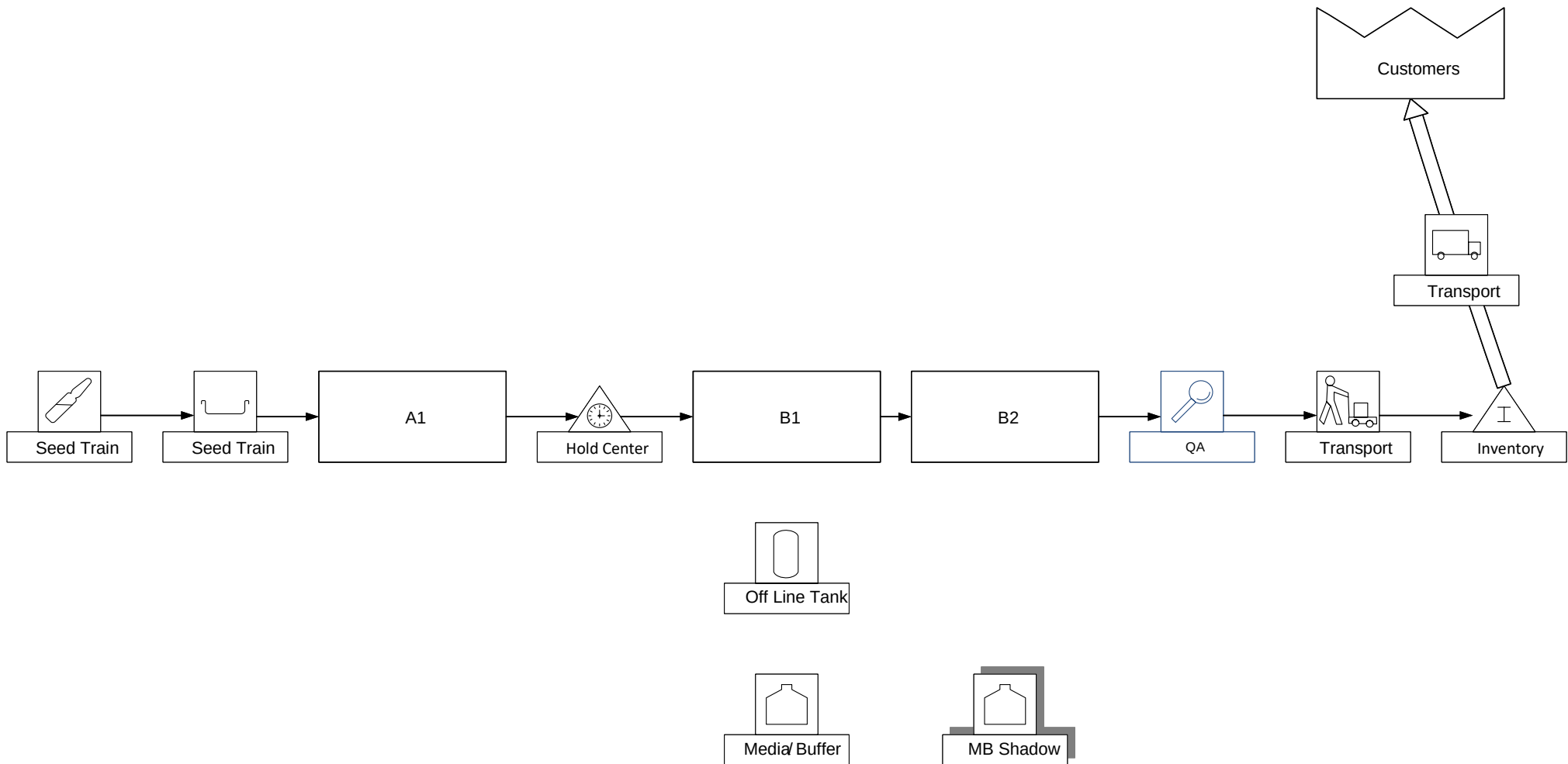
Capture the In-line Flow with Sketch Pharma

Use the Sketch Pharma stencil to draw the sketch shown in the image. Then connect them with Sequence arrows.



Material Flow for Next Exercise

In the next exercise you will draw the map below with the Sketch Pharma stencil. A printed copy of this page will help. Also, remember that all the pages in this course are available in the eL_QuickPharma.11Notes.pdf which was included in the ZIP file you downloaded for this course.



Sketch the Material Flow

With the “Sketch Pharma” draw the map shown on the previous page. Use sequence arrows to show the material flow.

For your convenience an image of the map from the previous page is available above.

For Online Course Only

- You learned:**
- How to open the Quick Pharma stencils and initialize a new map
 - How to quickly sketch the upstream, downstream, in-line and off-line material flow

Road Map:

eVSM Quick Pharma – Learning Road Map



Steps to sketch the process with Sketch Pharma

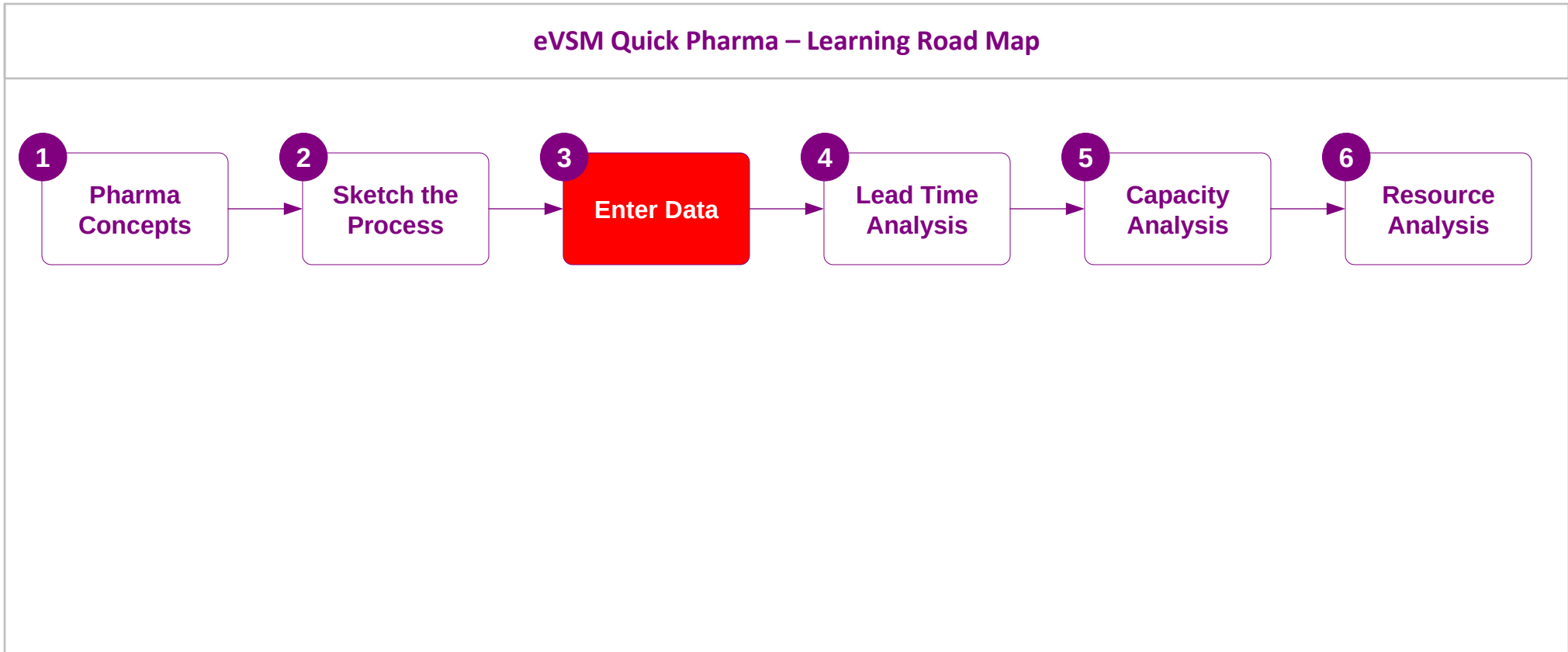
1. Open the Quick Pharma stencils and initiate the page
2. Drag out and line up the shapes from the Sketch Pharma stencil
3. Add Sequence arrows to show the flow of materials

What's next:

You will learn how to add data to the map, how to edit existing data, how to hide/show data, and how to import data via Excel.

Data Entry for Pharma value stream

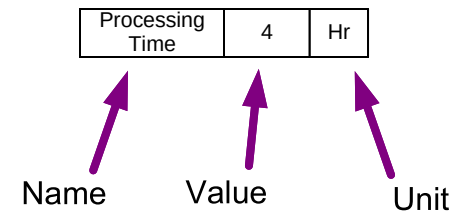
Data for the value stream is entered through some special data shapes which are attached to each activity. In this lesson you will learn how to add these data shapes automatically to a sketch map, how to work with these data shapes, and how additional data shapes can be used to perform optional extra calculations. You will also learn how to import data values through Excel.



Enter Data for the Pharma Process Flow

Working with Data on a Map

- Map data is stored in special data shapes which consist of a name, value, and unit.
- To change any field, double-click and then follow the on-screen instructions.
- To move or delete a data shape, you must select the value field.
- The Views (accessed with the  Views) button allows you to hide/show data shapes.
- All data shapes, including hidden ones, can be accessed through the  List Variables button. Select the Green center first, then click on the List Variables button.
- eVSM comes with a default list of variable names and units. New names and units can be added through “Name and Unit Manager” form which is accessed with the  NUM button.
- Default eVSM variable names and units should NOT be modified since they are used in the automated calculations.



The combination of a green shape along with all the data shapes attached is called a center.

Data shapes represent data for centers they are glued to.

These are VA and NVA (value added and non value added) data shapes. They are glued to centers via a flying glue point.

A0010	P	
In-Line Activity		
Utilization	Auto	%
Runs Per Batch	1	Run Btch
Processing Time	x.xx	Hr
Process Resource	0	Hr Btch

A0030		P
Transport		
Transport Time	x.xx	H

A0040	all	
Customers		
Customer Demand	x.xx	Run/Wk

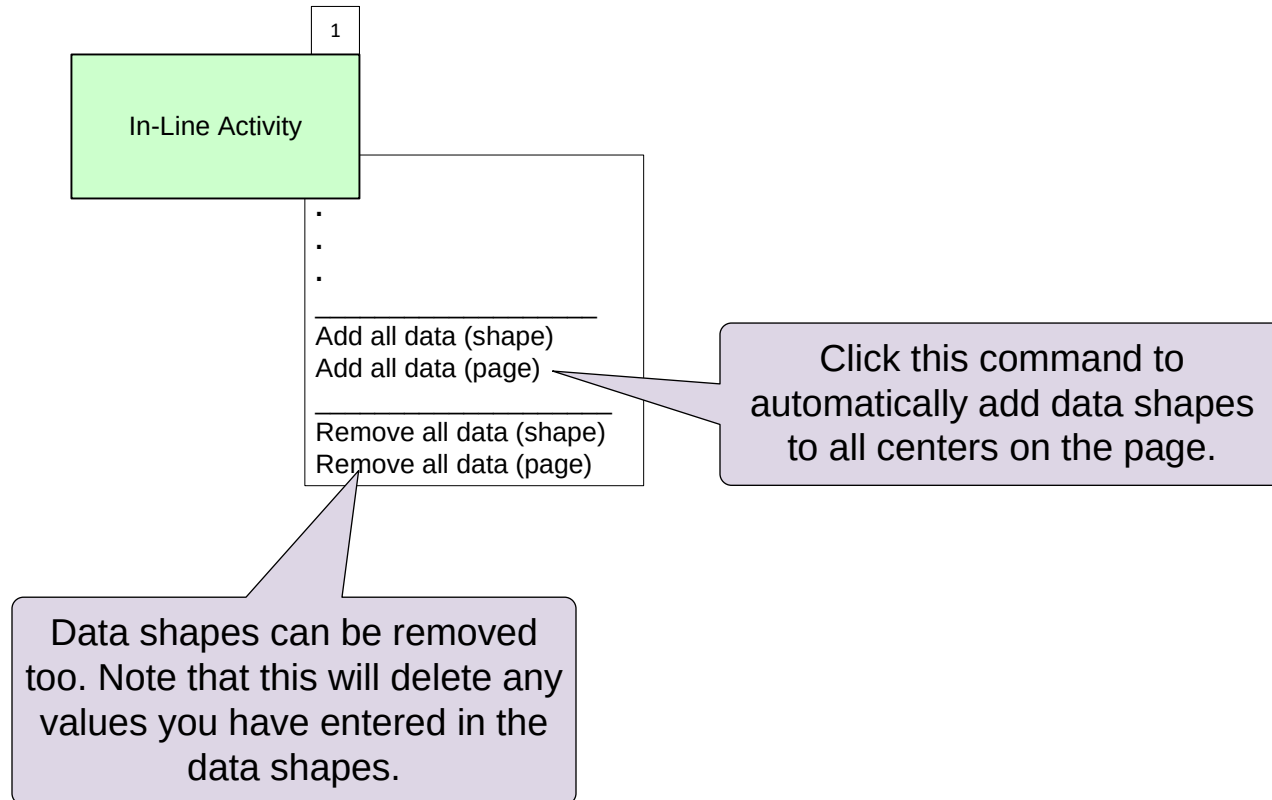
“xx” represents mandatory values you must provide. eVSM cannot perform calculations without these.

Blue values are automatically calculated by eVSM. Just leave these alone.

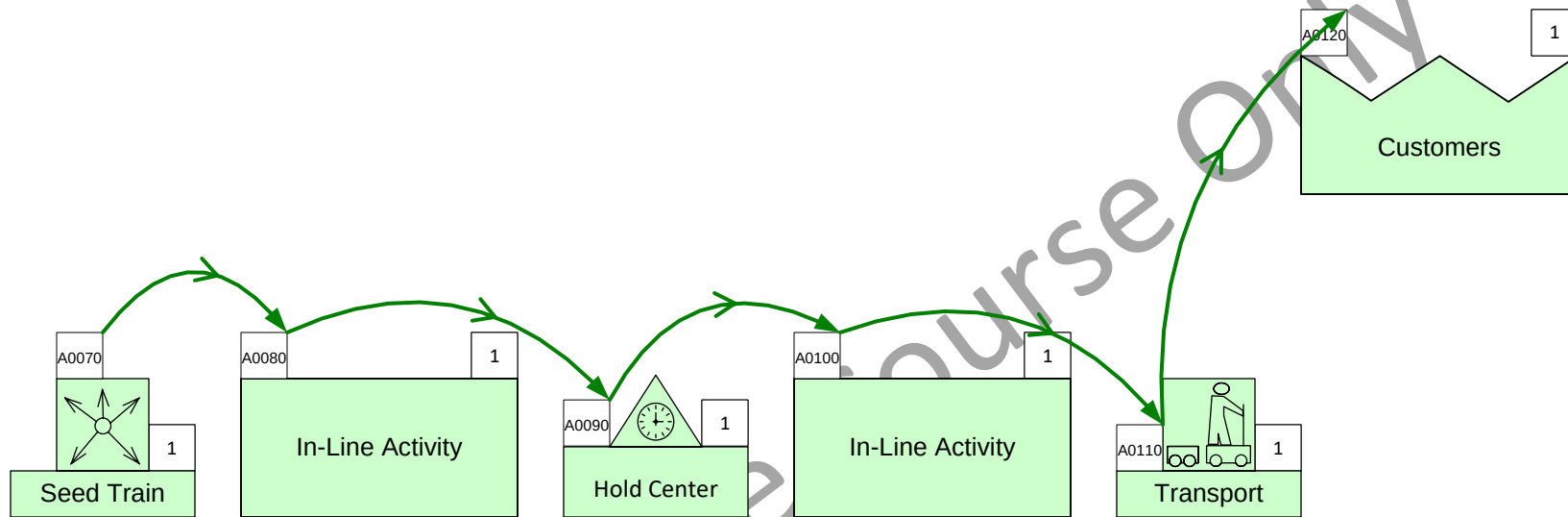
Auto	Hr
------	----

Data on Sketch Centers

Data shapes can be automatically added to a Sketch Pharma map. Just right-mouse click on any Sketch Pharma shape on the page.

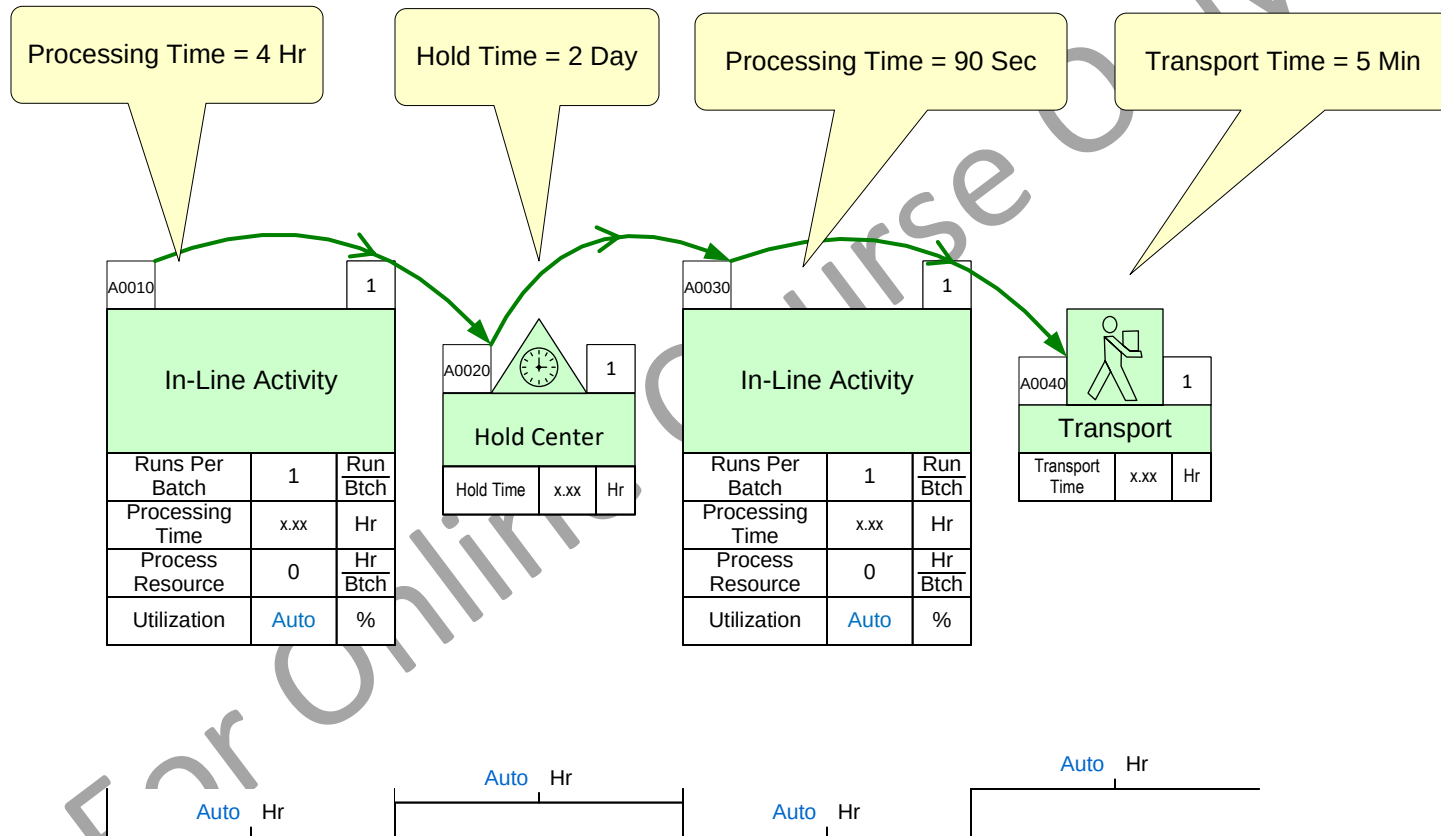


Add data shapes to this process flow. No need to enter data values or change units.



Units	Year	Wk	Wk	Shft
	52	7	21	8
	Wk	Day	Shft	Hr

Enter data values



40

Units	Year	Wk	Day	Day
	52	7	24	3
	Wk	Day	Hr	Shift

Working with Units

The default names and units can be seen in the NUM (Names and Units Manager) dialog. Additional names or units may be added if necessary.

Example: Add a new Unit called “Box”

The screenshot shows the 'Name and Unit Manager' dialog. A red 'XZY' icon labeled 'NUM' points to the dialog. The 'Map : Units' section shows a list of units with columns for 'Unit', 'On Map', 'US', and 'Metric'. A 'New Unit' dialog is open, showing the 'Select or Type In New Unit ID. Use 5 or less Alpha and Numeric Characters Only' field with 'Box' entered. The 'Alias' field is empty. The 'OK' button is highlighted. A blue dashed arrow points from the 'OK' button to the 'Unit' list in the 'Map : Units' section, with the text 'Click OK and the new unit will be seen the unit list'.

Name and Unit Manager

Name & Unit Sets

Save To Set..
Load From Set..
Delete Set..
Import Set..
Export Set..
Alias Mode: On Off
Export Alias Names Import Alias Names

Map : Units

Currency: \$

Unit	On Map	US	Metric
%	No		
0or1	No		
Btch	No		
Day	No		
Hr	Yes		
Kit	No		
Load	No		
Min	Yes		
none	No		
Run	No		

New Unit
Select or Type In New Unit ID. Use 5 or less Alpha and Numeric Characters Only
Box
Alias
Cancel OK

Map : Names (NVU's)

Name	On Map	Hidden	Default Unit	Filter:
Activity Time	No	No	Hr/Wk	
Adjacent Batch Resource	No	No	Hr/Btch	
Adjacent Lead Time	No	No	Day	
Adjacent Prd Resource	No	No	Hr/Wk	
After Post Hold	No	No	Hr	
Allocated Resource	No	No	Staff	
Batch Record Resource	No	No	Hr/Btch	
Capacity	No	Yes	Run/Wk	
CIP Resource	No	No	Hr/Btch	
CIP Time	No	No	Hr	

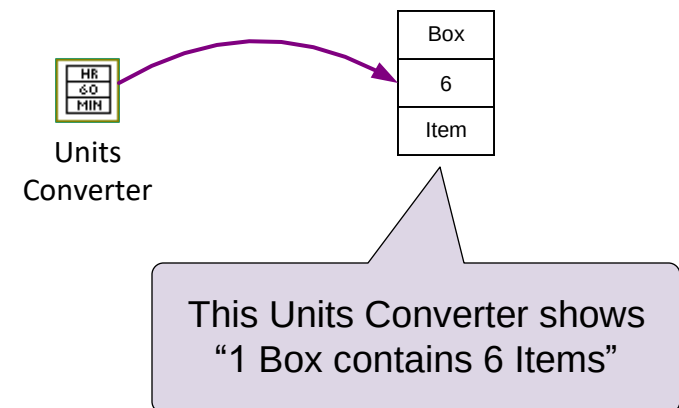
Load From Map Source/Target Pages Equation Manager.. OK

Units Families

Units can be organized into families such as times, weight, length, etc. This allows you to enter data (or view calculations results), and use different units on different parts of the map. For example, you may want to show quantities in Pallets in one place and Boxes in another place.

Some families have already been set up for you, e.g. Seconds, minutes, hours, days, or all in one family.

Units are associated to families with Units Converter shape.



Add new unit convertor

Create a new unit “Tray” and then use a Units Converter to show that there are 12 “Kit” per “Tray”.

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Units	Year	Wk	Wk	Shft
	52	7	21	8
	Wk	Day	Shft	Hr

Hide/Show Data Shapes

The Quick Pharma centers include data shapes for data input and to display calculated values. By default, most of these data shapes are hidden to keep the map view simple. Visibility of these is controlled through the Views form.

Views Form



Variable Visibility

Center/Add-on Name	Variable Name(s)	Visibility	Show in List Vars
+ [Icon] Customer Center		☐ All	☒ All
+ [Icon] In Line Adjacent			
+ [Icon] In Line Seed Train			
- [Icon] In Line Activity	- Capacity	☐	☒
	Computed Time	☐	☒
	Demand	☐	☒
	Demand %	☐	☒
	Equipment Time	☐	☒
	Lead Time	☐	☒
	Process Resource	☒	☒
	Processing Time	☒	☒
	Resource Required	☐	☒
	Runs Per Batch	☐	☒

Default Variable Visibilit Hide Auto Variables Cancel OK

Use this checkbox to hide/show ALL data shapes on the map

This column of switches is used to determine which variables will appear in the List Variables form (covered in a later page)

Use Expand/Collapse button to access all data shapes for a center

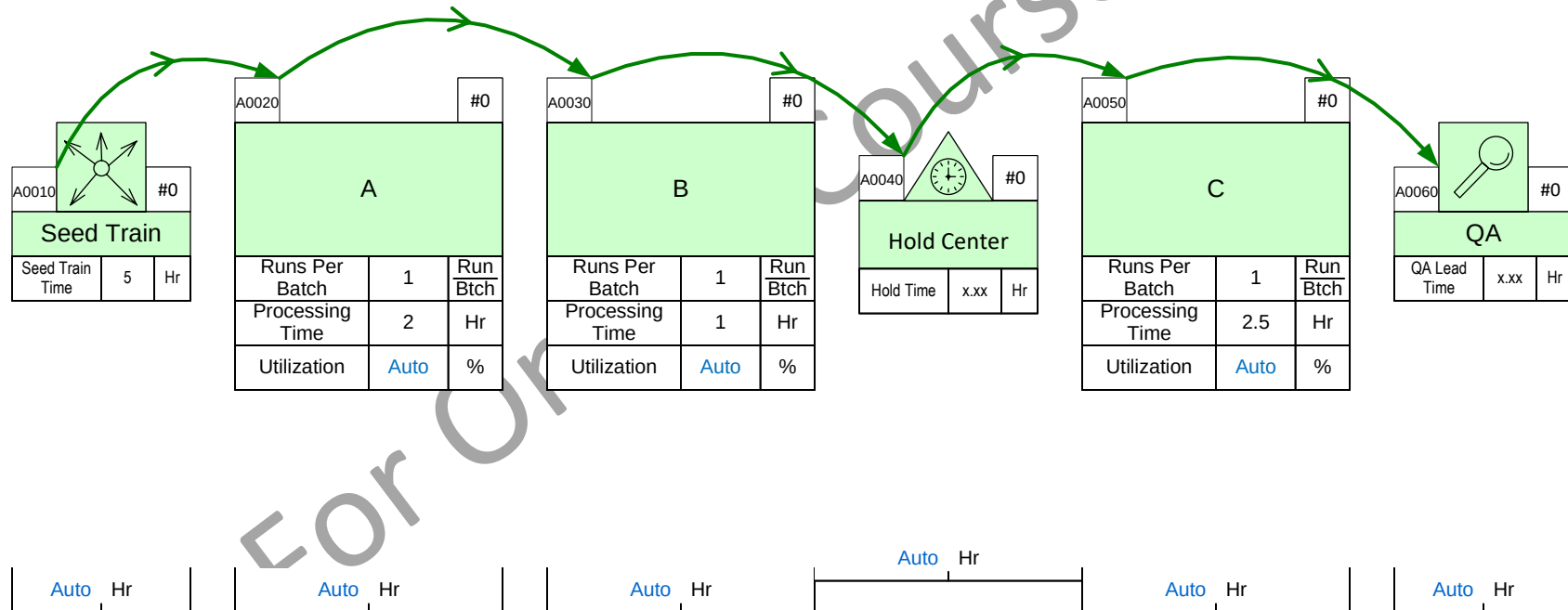
These switches control which data shapes will be visible on the map

This button reverts the switches back to the default state (as it was for a new map)

Use this button to hide all data shapes containing calculated values

Hide/Show Data with Views Dialog

1. Hide the time line (value added and non-value added shapes)
2. Make "Process Resource" data shapes visible



Units	Year	Wk	Wk	Shift
	52	7	21	8
	Wk	Day	Shift	Hr

Using the Yellow Add-ons in Quick Stencils

Many common calculations are built into the main centers (green shapes) and sequence arrows. The yellow icons in the Quick stencils provide additional modifications and calculations.

How to Use Add-ons

Just drag out the desired add-on from the stencil and glue it to the bottom of the data shapes stack. New add-ons can also be dropped on the green center shape and it will automatically get attached to the bottom of the stack.

Example

This plant works 21 shifts per week as indicated in the Time Center. Therefore it is assumed the packaging process is available 21 Shft/Wk. However, what if the packaging process is required 14 shifts per week? You can use the Activity Time yellow add-on to show the actual available time.

Time Center

Units	Year	Wk	Wk	Shft
	52	7	21	8
	Wk	Day	Shft	Hr

Plant production
21 shifts per week



Activity Time

Packaging operation
production time per
week

A0040			P
Packaging			
Utilization	Auto	%	
Runs Per Batch	1	Run Btch	
Processing Time	4	Hr	
Process Resource	0	Hr Btch	
Activity Time	14	Shft Wk	
Auto Hr			

Add-ons Icon Positions in the Stencil

There is a unique correspondence between green icons in the stencil and the yellow icons that immediately follow the green icon. ONLY with the green icons they “belong” to.

Units	Year	Wk	Wk	Shft
	52	7	21	8
	Wk	Day	Shft	Hr

Add-ons for In-Line Activity Center



In Line Activity

A0050			P
In-Line Activity			
Utilization	Auto	%	
Runs Per Batch	1	Run/Btch	
Processing Time	x.xx	Hr	
Process Resource	0	Hr/Btch	

This center is used for both upstream and downstream activities. For upstream usage you may be making a batch from which several downstream runs are made. Hence the runs per batch variable.

The **Processing Time** is considered as part of the lead time.



In Proc Hold

In Process Hold Time	xx	Hr
In Process Hold Resource	xx	Hr/Btch

The **In Process Hold Time** is considered as part of the lead time.



Activity Pre

Pre Time	0	Hr
Pre Resource	0	Hr/Btch

The **Pre Time** is considered as off line and just before processing begins.



Activity Post

Post Time	x.xx	Hr
Post Resource	0	Hr/Btch

The **Post Time** is considered as off line and just after processing completes. It can be a caustic clean ahead of CIP and SIP.



After Post Hold

After Post Hold	xx	Hr
-----------------	----	----

The **After Post Hold** is considered as off line and in parallel with CIP and SIP. So the related equipment time is $\text{Max}(\text{After Post Hold}, \text{CIP} + \text{SIP})$.
CIP (Clean In Place)



CIP

CIP Time	xx	Hr
CIP Resource	xx	Hr/Btch



SIP

SIP Time	xx	Hr
SIP Resource	xx	Hr/Btch

SIP (Steam In Place)



Turn Around

Turnaround Time	xx	Hr
Turnaround Resource	xx	Hr/Btch

This can be used instead of the detailed breakup of the non-process times.



Batch Record

Batch Record Resource	0	Hr/Btch
-----------------------	---	---------

The recording is assumed off line and not to impact lead time.



Manual Time

Manual Time	0	Hr
-------------	---	----

This is not used for analytics but useful to consider opportunity for resource optimization.



Activity Stations

Stations	1	Stn
----------	---	-----

This is used for duplicate stations that are run in parallel.



Activity RFT

Right First Time	100	%
------------------	-----	---

Not used for analytics but useful for considering improvements.



Yield

Step Yield	xx	%
------------	----	---

Not used for analytics but useful for considering improvements.



Downtime

Downtime	xx	%
----------	----	---

Equipment downtime. Used to estimate effective capacity.



Activity Time

Activity Time	x.xx	Hr/Wk
---------------	------	-------

Activity time (after breaks) used if different at a station from that of the overall plant.

Quick Pharma Add-ons Examples

Example 1

Activity Ds2 has 3 Stations working in parallel.



Activity Stations

A0060			#0
Ds2			
Runs Per Batch	1	Run Btch	
Processing Time	4	Hr	
Process Resource	2	Hr Btch	
Stations	3	Stn	

Auto Hr

Example 2

The production line works 3 shifts a day, but the Packaging activity is available only for 1 shift per day.



Activity Time

A0070			#0
Packaging			
Runs Per Batch	1	Run Btch	
Processing Time	4	Hr	
Process Resource	2	Hr Btch	
Activity Time	1	Hr Day	

Auto Hr

Example 3

We have significant Pre and Post time.



Activity Pre



Activity Post

A0080			#0
Ds2			
Runs Per Batch	1	Run Btch	
Processing Time	4	Hr	
Process Resource	2	Hr Btch	
Pre Time	2	Hr	
Pre Resource	2	Hr Btch	
Post Time	1.5	Hr	
Post Resource	1	Hr Btch	

Auto Hr

Example 4

The equipment at activity Us2 has a downtime of 15% and a yield of 80%.



Downtime



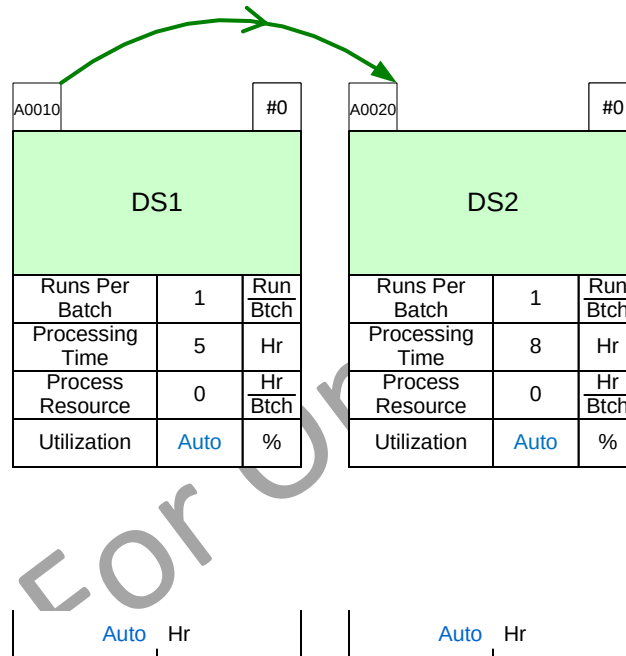
Yield

A0090			#0
Us2			
Runs Per Batch	1	Run	Btch
Processing Time	4	Hr	
Process Resource	2	Hr	Btch
Step Yield	80	%	
Downtime	15	%	

Auto Hr

Use add-ons to represent the following in the two activity centers

1. Activity DS1 is available 15 shifts per week
2. Activity DS2 has 3 stations



Plant Operating Hours

Units	Year	Wk	Wk	Shft
	52	7	21	8
	Wk	Day	Shft	Hr

Edit Hidden Data with List Variables

Data shapes can be edited directly when they are visible on the map. The List Variables form provides a way of viewing and editing values for hidden data shapes.

1 Select the center you wish to edit or view

2 Click the List Variables button

3 View values here. Double-click a value to edit

The 'View Center Data' dialog box for 'In-Line Activity' A0030 displays the following variables and values:

Variable	Value
Process Resource	0 Hr/Btch
Processing Time	x.xx Hr
Runs Per Batch	1 Run/Btch
Utilization	Auto %
Value Added	Auto Hr
Capacity	Auto Run/Wk
Computed Time	Auto Hr/Wk
Demand	Auto Run/Wk
Demand %	100 %
Equipment Time	Auto Hr/Btch
Lead Time	Auto Hr
Resource Required	Auto Hr/Wk

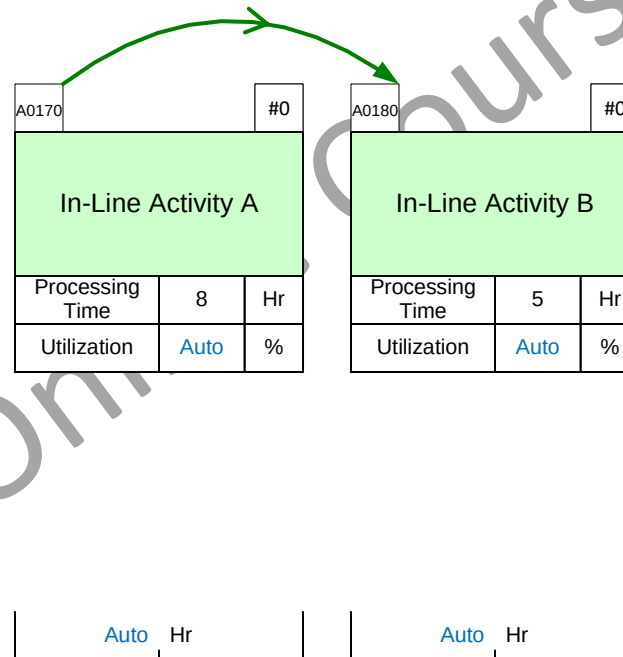
☐ View All Variables
Double-Click Variables to Edit

Viewing and Editing Data with List Variable

Use the List Variables dialog to set the values for

Process Resource = 5 Hr/Btch

for both activities below.



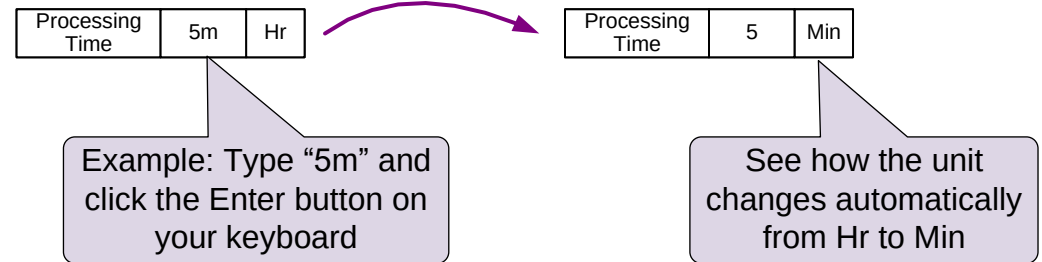
Units	Year	Wk	Wk	Shft
	52	7	21	8
	Wk	Day	Shft	Hr

Refresher Tips

Topics below were covered in more detail in the Time Mapping course. The notes here serve as a refresher.

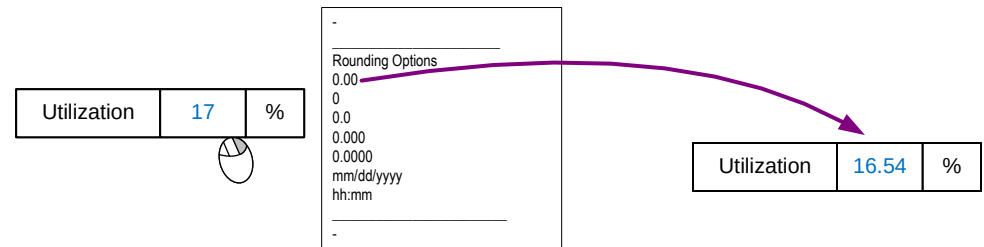
1. Instant change of Units

Sometimes you need to use different time units for the same variable on different parts of the map. E.g. Processing Time may be in seconds at some activities and hours at others. You can of course double-click the unit field and change the unit where required.



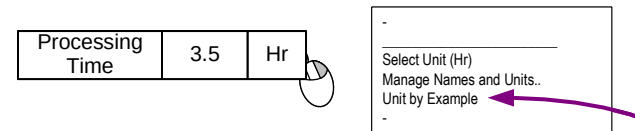
2. Rounding

You can control the decimal places displayed in data shapes with the Rounding command in the right-mouse button menu.



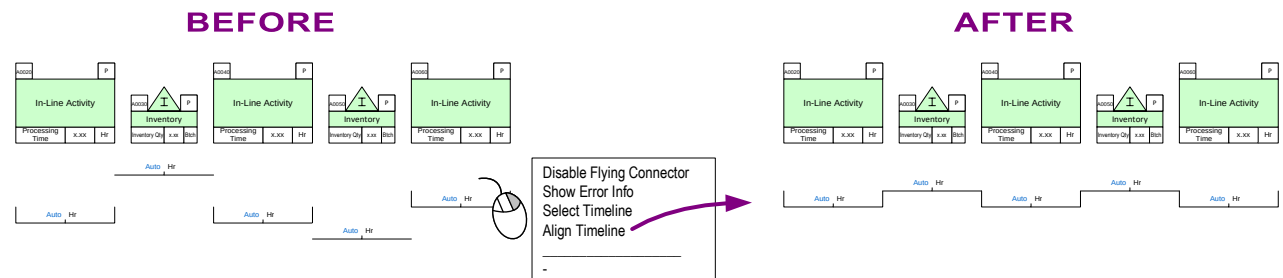
3. Units by Example

When you need to set the unit for a variable throughout the whole map, you can set it on one data shape, then use the “Units by Example” command to change it everywhere.

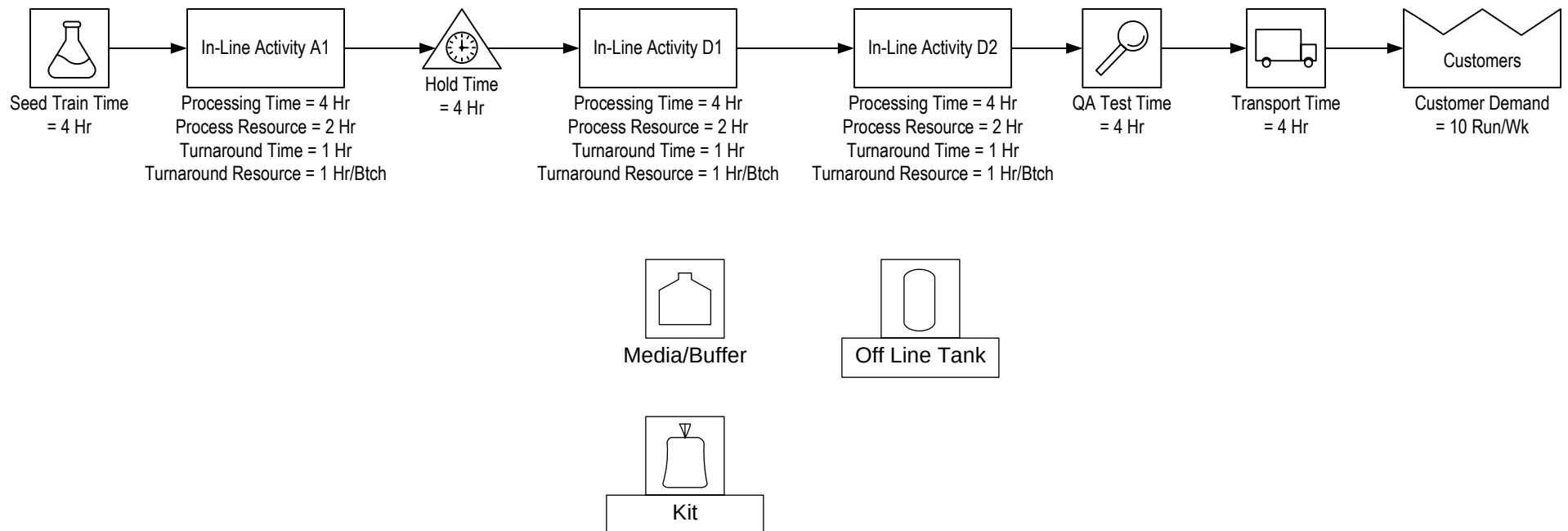


4. Align Timeline

The timeline (value added and non-value added data shapes) can be quickly aligned with the “Align Timeline” command.



Map for Next Exercise



Draw the map shown on the previous page, complete with all data shown, and sequence arrows.

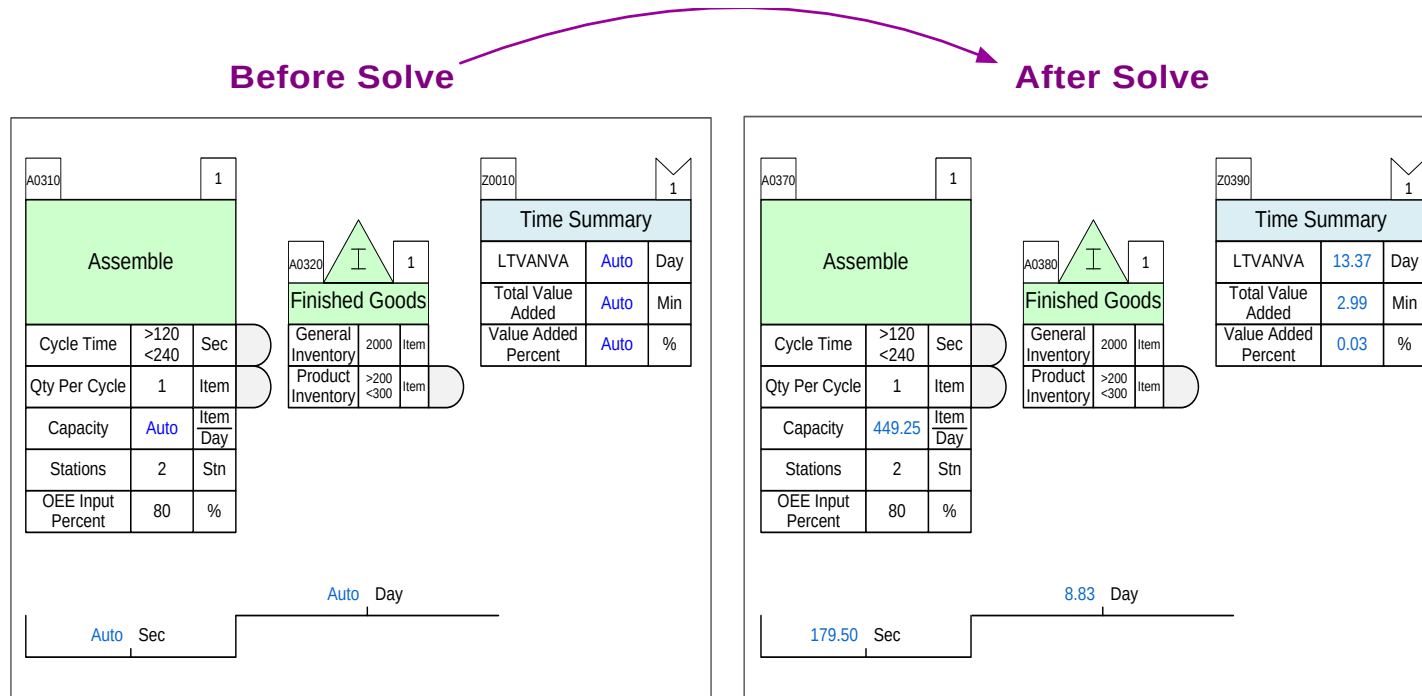
An image of the map is available here.

For Online Course Only

Units	Year	Wk	Wk	Shft
	52	7	21	8
53	Wk	Day	Shft	Hr

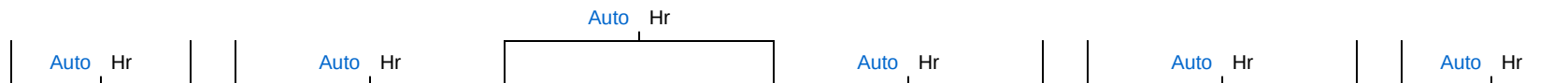
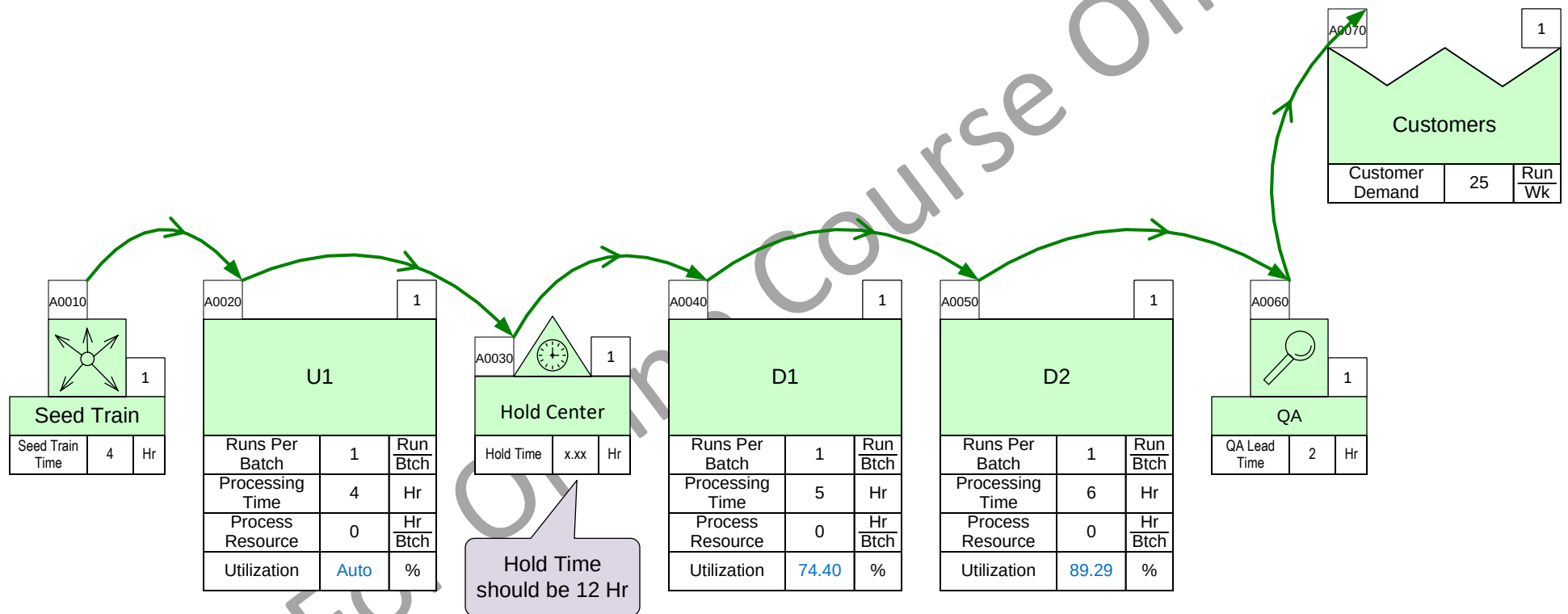
Solve the Map

When ready, click the  **Solve** button. This will first check the map for any problems (e.g. missing data, disconnected arrows, duplicate tag numbers, etc.). If there problems exist, it will stop the Solver and report the problems. Otherwise, it will complete the calculations, and the blue values on the map will show the results. Solve also updates any charts on the map.



Solve this map

Just hit the Solve button, fix any errors reported, and Solve again.



55

Units	Year	Wk	Wk	Shift
	52	7	21	8
	Wk	Day	Shift	Hr

Data Input through Excel

Sometimes it is more convenient or easier to input data through Excel rather than type directly on the map. eVSM facilitates this with:



Create XL - Creates an Excel file which represents all the data input values for the current map.



Import XL - Import XL pulls the data in from Excel to the map.

The columns represent the variables

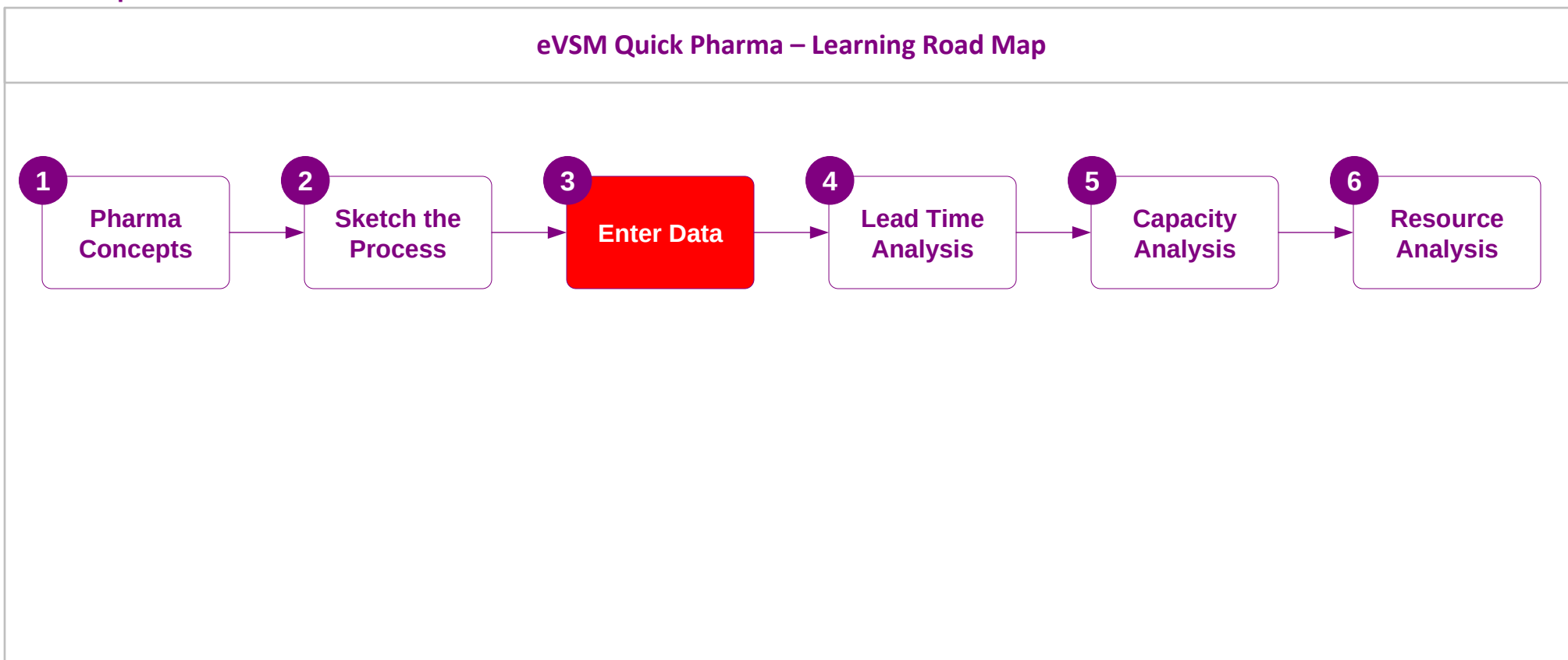
sent map

	B	C	D	E	F	G	H	I	J		L	M	N	O	P	Q	
1	Tag	Center Type	Operation	Data	Data	Data	Data	Data	Data		Data	Data	Data	Data	Data	Data	
				Adjacent Lead Time	Batch Record Resource	Customer Demand	Demand %	Hold Time	Inventory Qty		Pre Resource	Pre Time	Process Resource	Processing Time	QA Lead Time	QA Resource	Run Scrap
				Day	Min/Btch	Run/Wk	%	Day	Btch		Hr/Btch	Hr	Hr/Btch	Hr	Hr	Hr/Btch	Run/Year
4																	
5	A0010	In Line Seed Train	St3														
6	A0030	In Line Activity	Up1		x.xx		100.0%						0 x.xx				
7	A0040	In Line Activity	Up2		x.xx		100.0%						0 x.xx				
8	A0050	In Line Activity	Up3		x.xx		100.0%				1 x.xx		0 x.xx				
9	A0060	In Line Hold	Hold Center					x.xx									
10	A0070	In Line Activity	Ds1				100.0%				0 x.xx		0 x.xx				
11	A0080	In Line Activity	Ds2				100.0%				0 x.xx		0 x.xx				
12	A0090	In Line QA	QA												x.xx	1	
13	A0100	Customer Center	Customers			x.xx											0
14	Z0360	Capacity Summary	Capacity Summary														
15																	
16		Add Center..															
17		Add Center..															
18		Add Center..															

Th

- You learned:**
- How to work with data shapes
 - How to automatically add data shapes to an eVSM sketch map
 - How to work with eVSM Add-ons for optional calculations
 - How to import data values through Excel

Road Map:

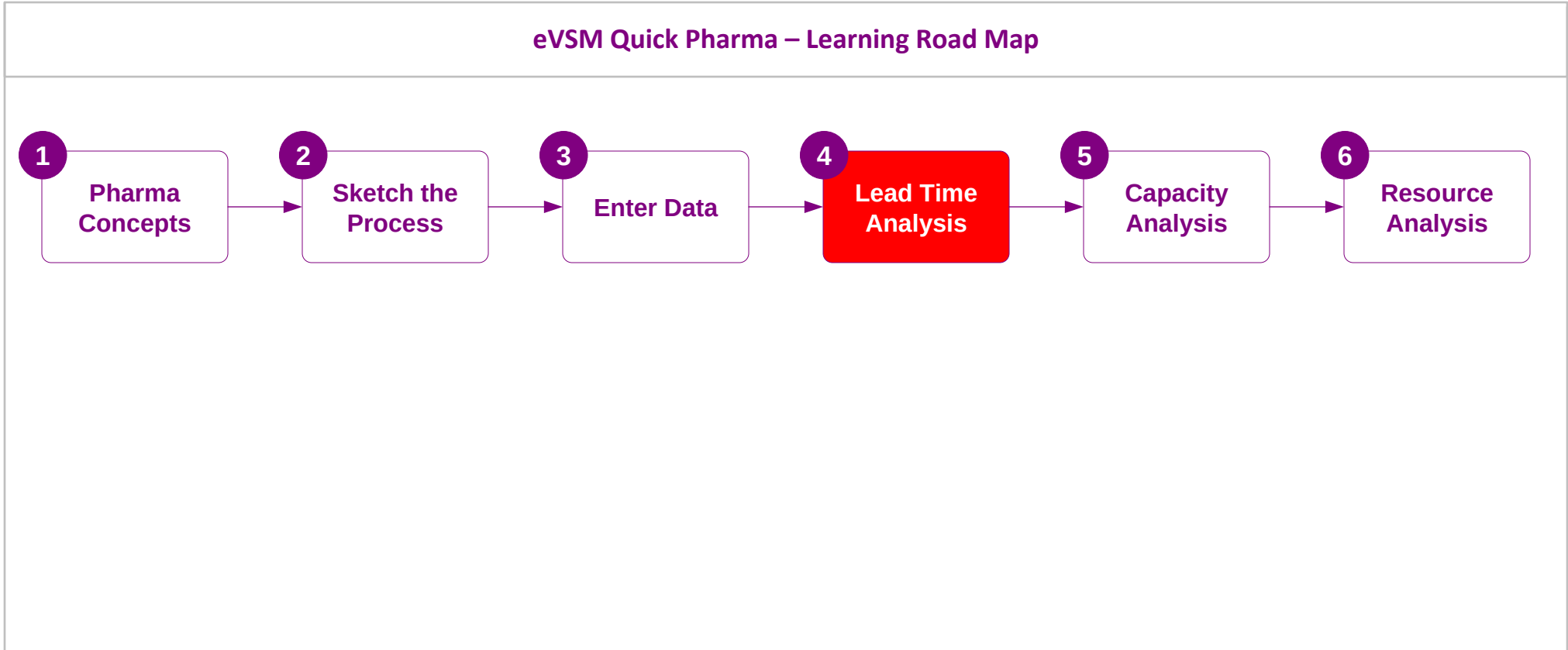


What's next:

How to calculate the and visualize the lead time for value streams,

Lead Time Analysis

In lesson 1, you learned the concepts and components of lead time. This this lesson you will learn how to input data for lead time analysis and then see the calculations in a summary and a chart.



Lead time analysis for the Pharma Value Stream

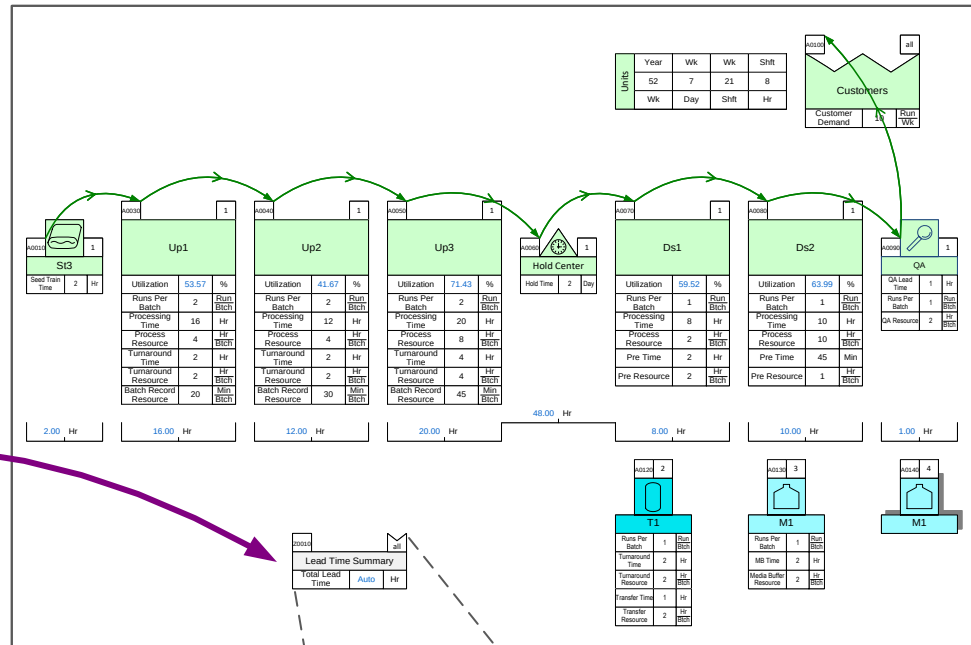
Time Summary

The Lead Time Summary center calculates the total time for the materials to go through the full in-line process.

1 Drop the Lead Time Summary center on the map



Lead Time Summary

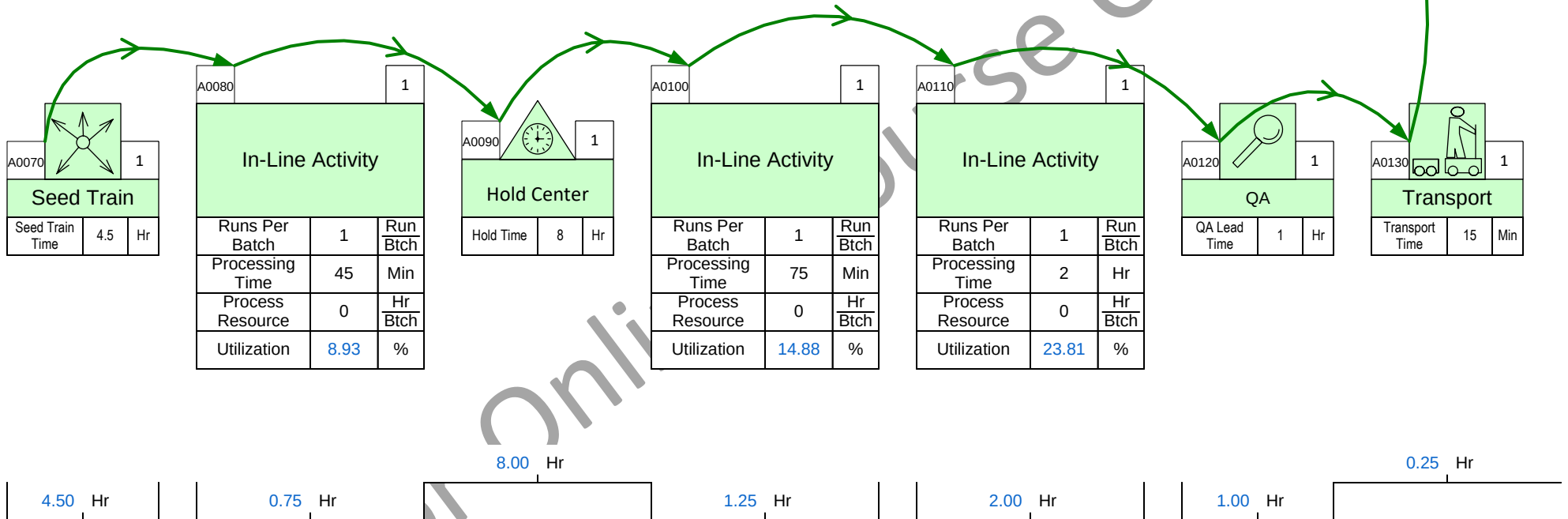


2 Solve and see the result

Lead Time Summary		
Total Lead Time	37.2	Hr

Add a Time Summary and Solve the Model to calculate the total lead time for the in-line processes.

Units	Year	Wk	Wk	Shft
	52	7	21	8
	Wk	Day	Shft	Hr



Lead Time Chart

The Lead Time Chart visualizes the lead time components.

1 Drop the Lead Time Chart shape on the page.



Lead Time Chart

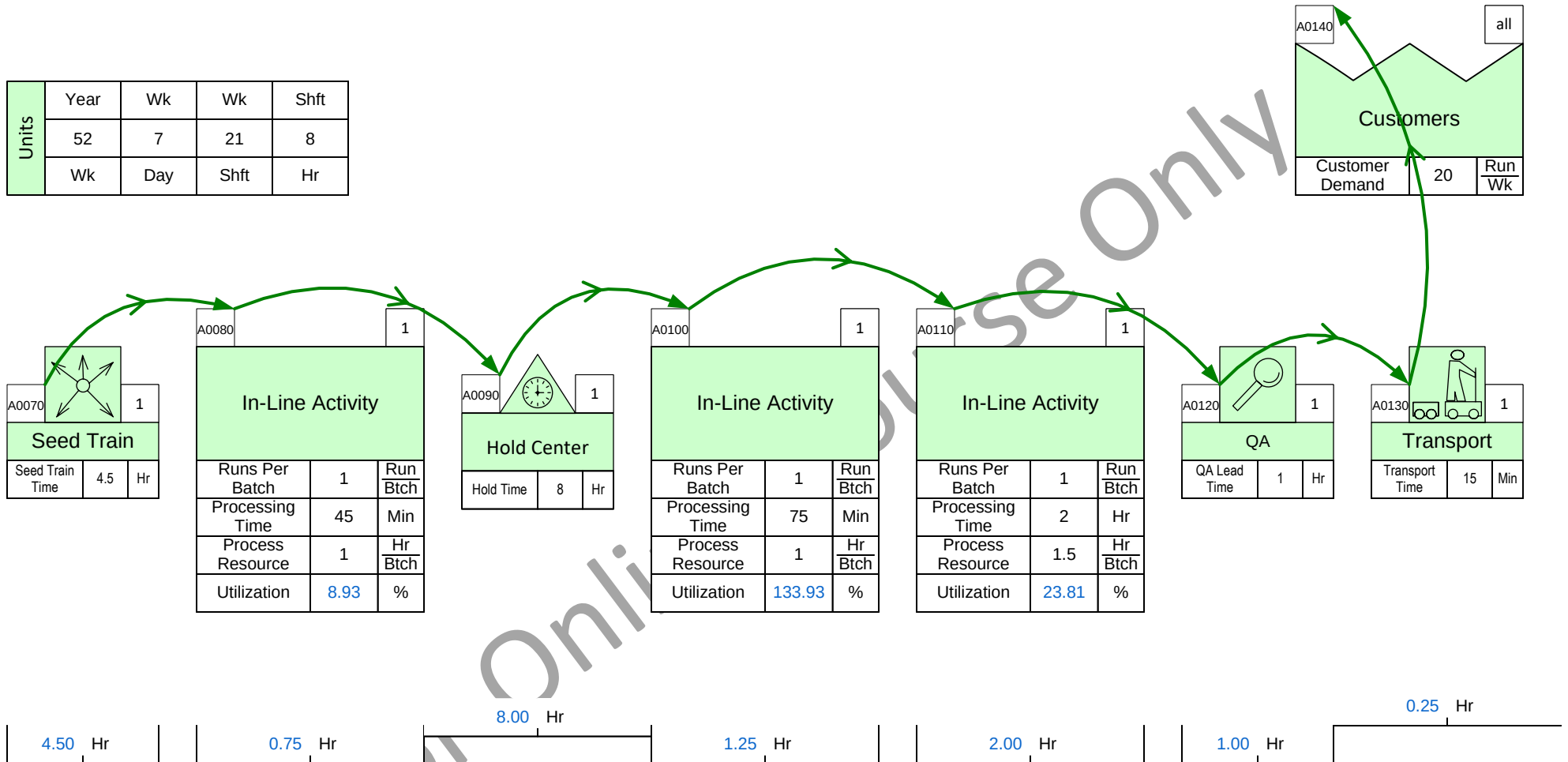
2 In the right-mouse menu of the chart shape, select the "Plot Lead Time" command.

The right-mouse menu of the chart shape also includes "Edit Ladder Chart Data" command which allows you to change units, scale, etc.



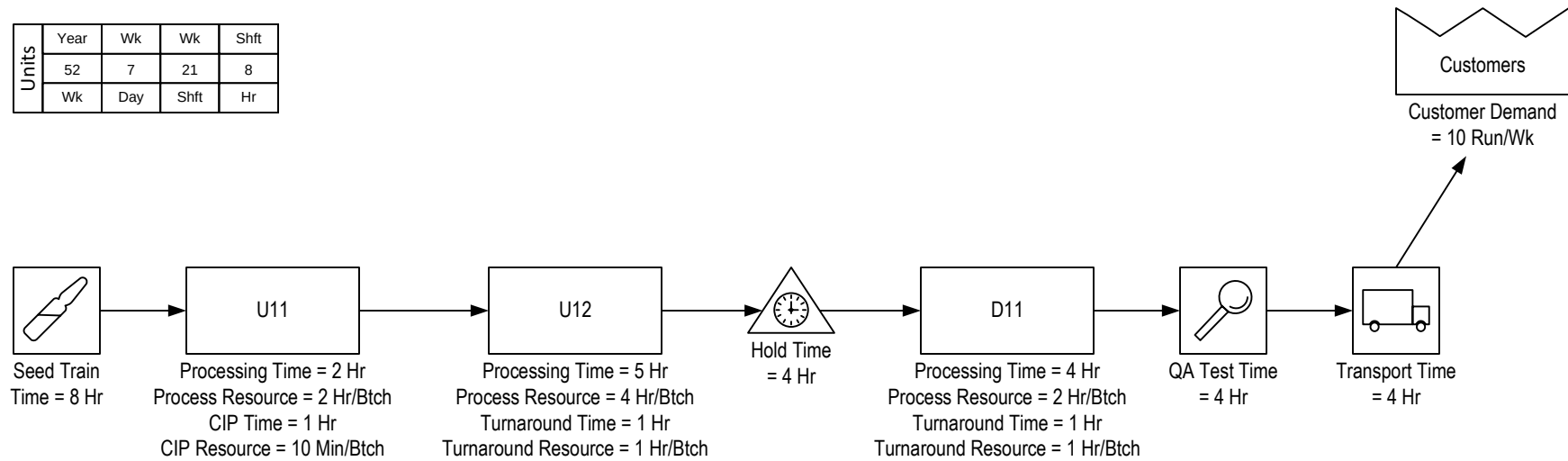
Plot the Lead Time chart for this value stream

Units	Year	Wk	Wk	Shft
	52	7	21	8
	Wk	Day	Shft	Hr



Map for Next Exercise

Units	Year	Wk	Wk	Shift
	52	7	21	8
	Wk	Day	Shift	Hr



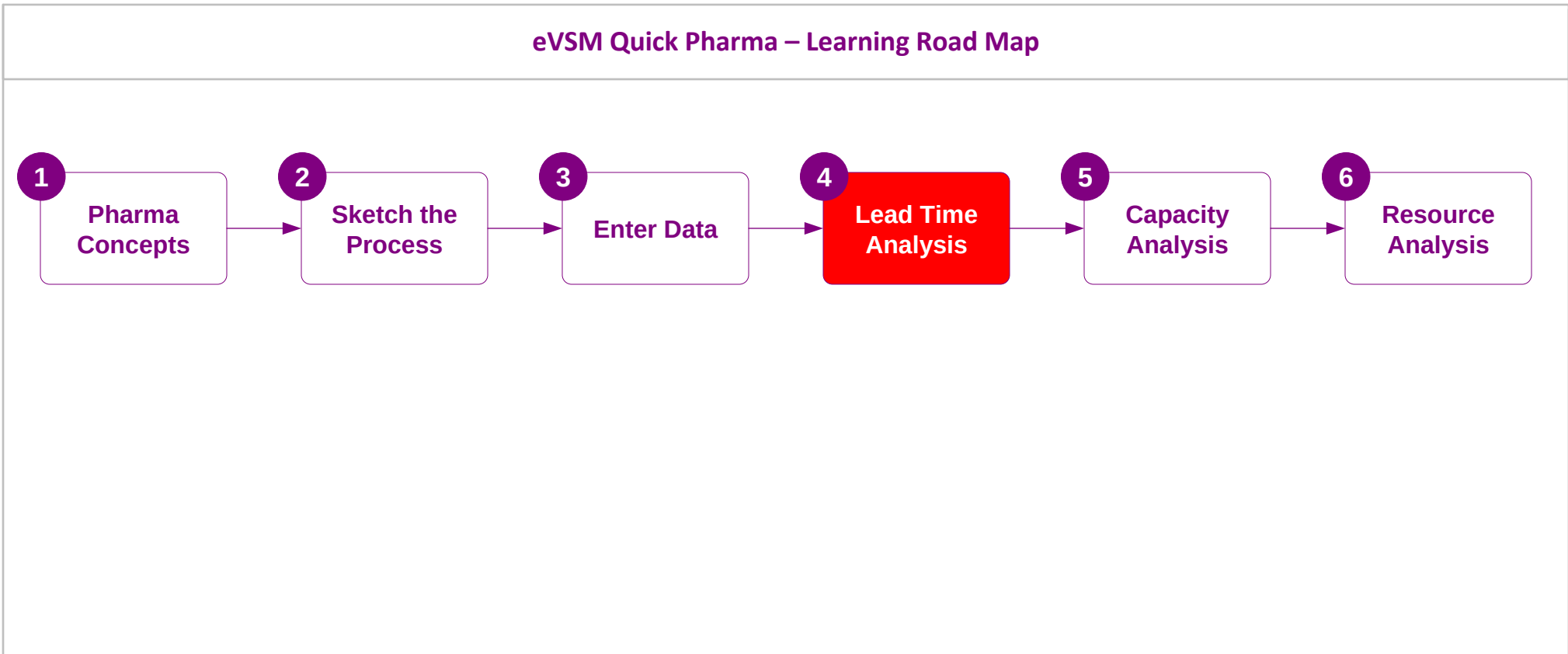
Draw the map shown on the previous page

Image of the map is
available here

For Online Course Only

- You learned:**
- How to use a lead time summary center.
 - How to plot a lead time chart.
 - You saw where the controls are to customize the chart for scale, tweak units, title, legend, etc.

Road Map:

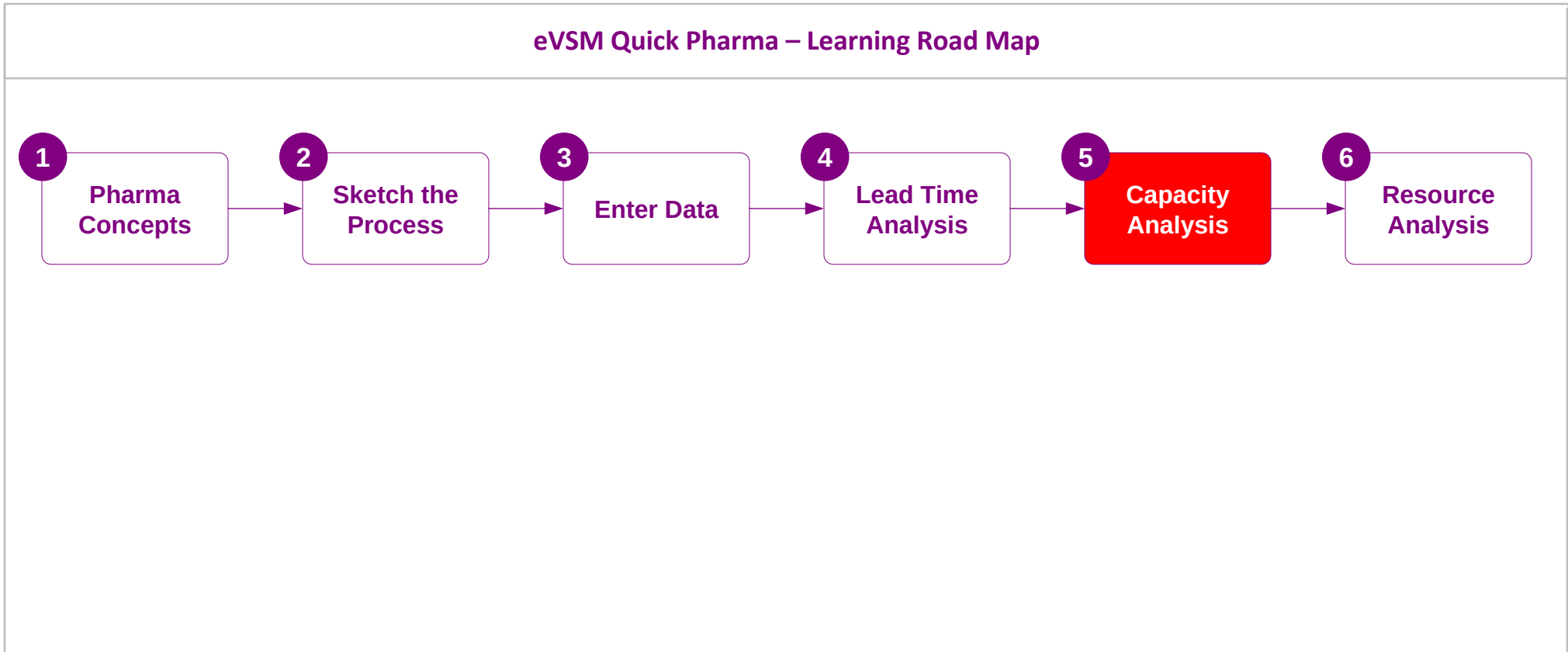


What's next:

You will learn about capacity and capacity analytics in more detail.

Capacity Analysis

In this lesson we will learn how to analyze the capacity of the value stream. You will see how to identify any bottle-necks, how to calculate the line capacity, how to represent causes of capacity loss, and how to explore impact on capacity of improvement ideas in eVSM.



Capacity analysis for the Pharma Process Flow

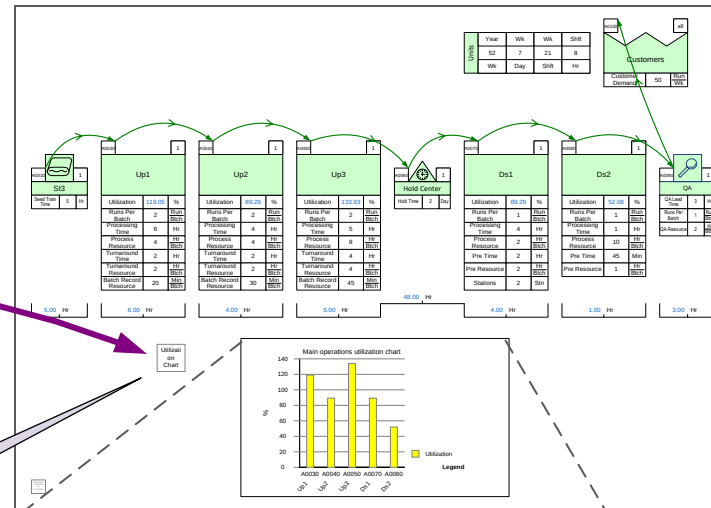
Capacity Analysis

The Utilization Chart in the Quick Pharma stencil summarizes the capacity utilization per activity for the whole value stream. Steps to plot the Utilization chart:

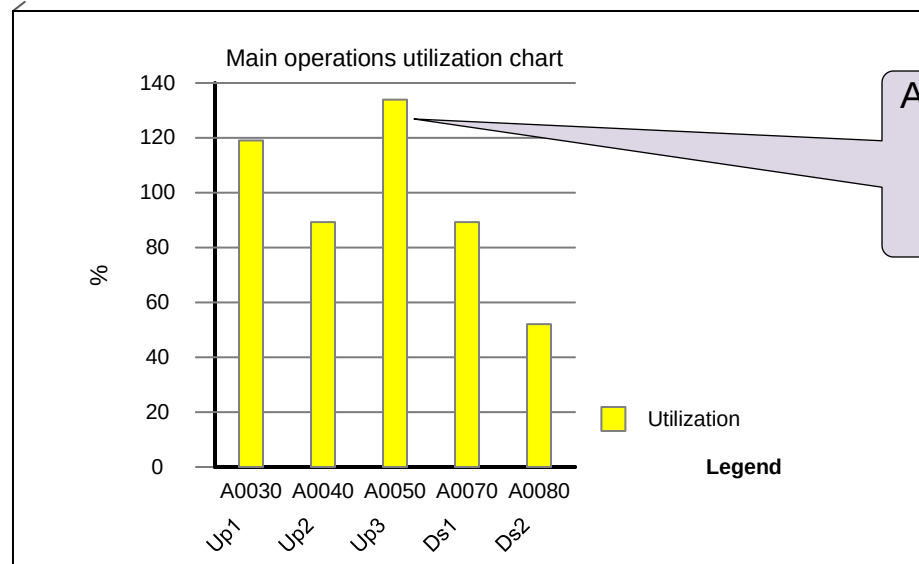
- 1 After the map has been solved, drop the Utilization Chart shape on the map



Utilization Chart



- 2 Click the “Plot Chart” command in the right-mouse menu



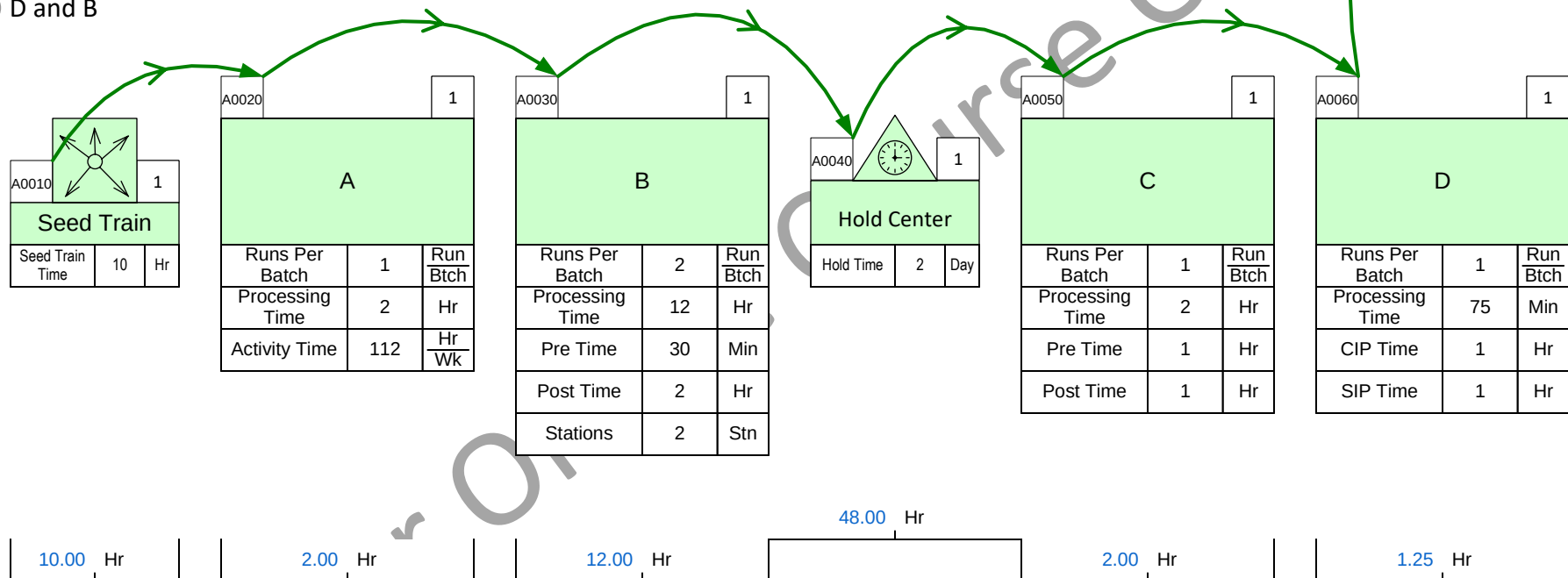
Activities which exceed 100% utilization will have trouble meeting customer demand

Which activities are not able to meet customer demand?

- ☐ A and B
- ☐ B and C
- ☐ A and D
- ☐ D and B

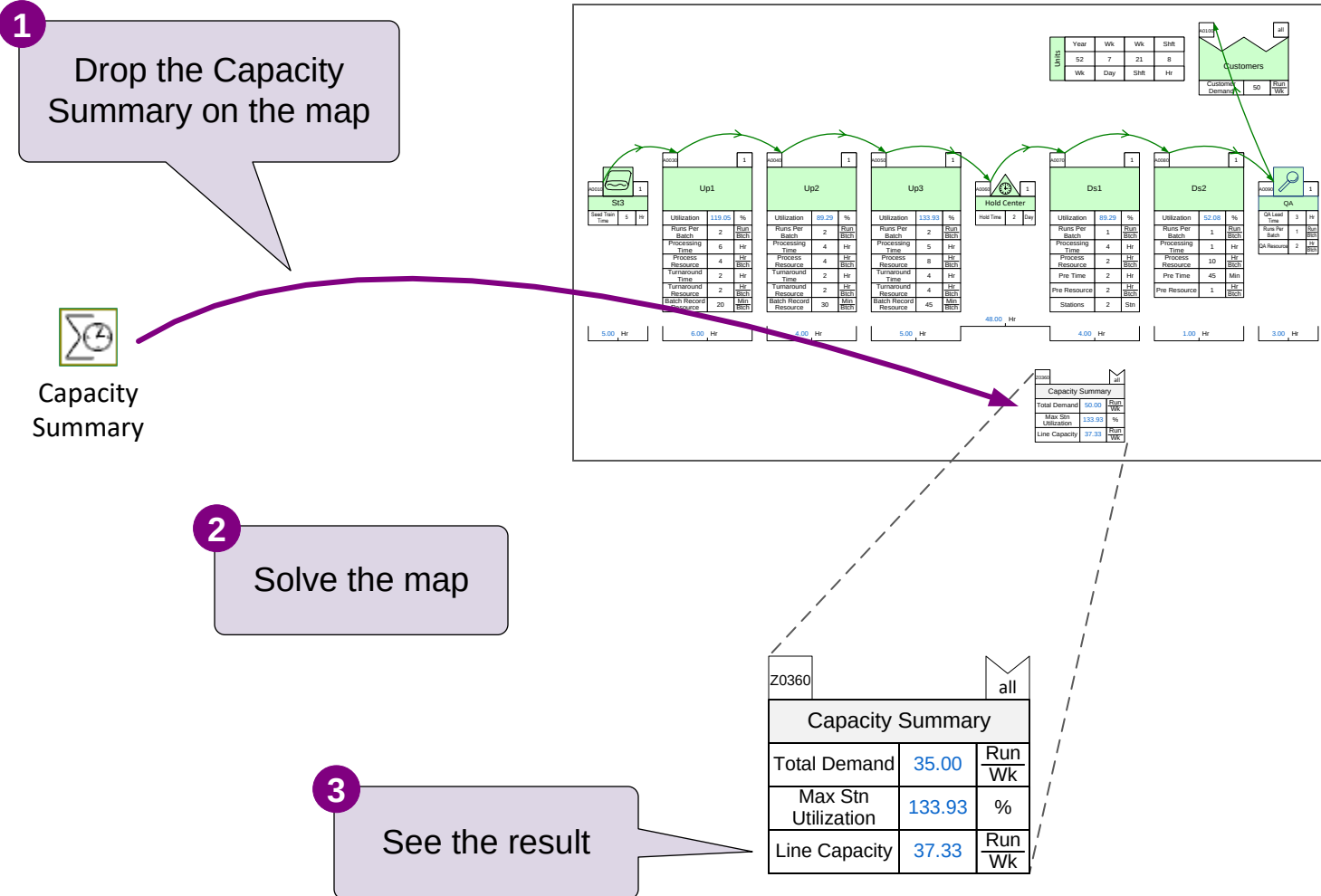
Units	Year	Wk	Wk	Shft
	52	7	21	8
	Wk	Day	Shft	Hr

A0070	all	
Customers		
Customer Demand	50	Run Wk



Line Capacity

The line capacity is calculated in Capacity Summary. Steps to use:



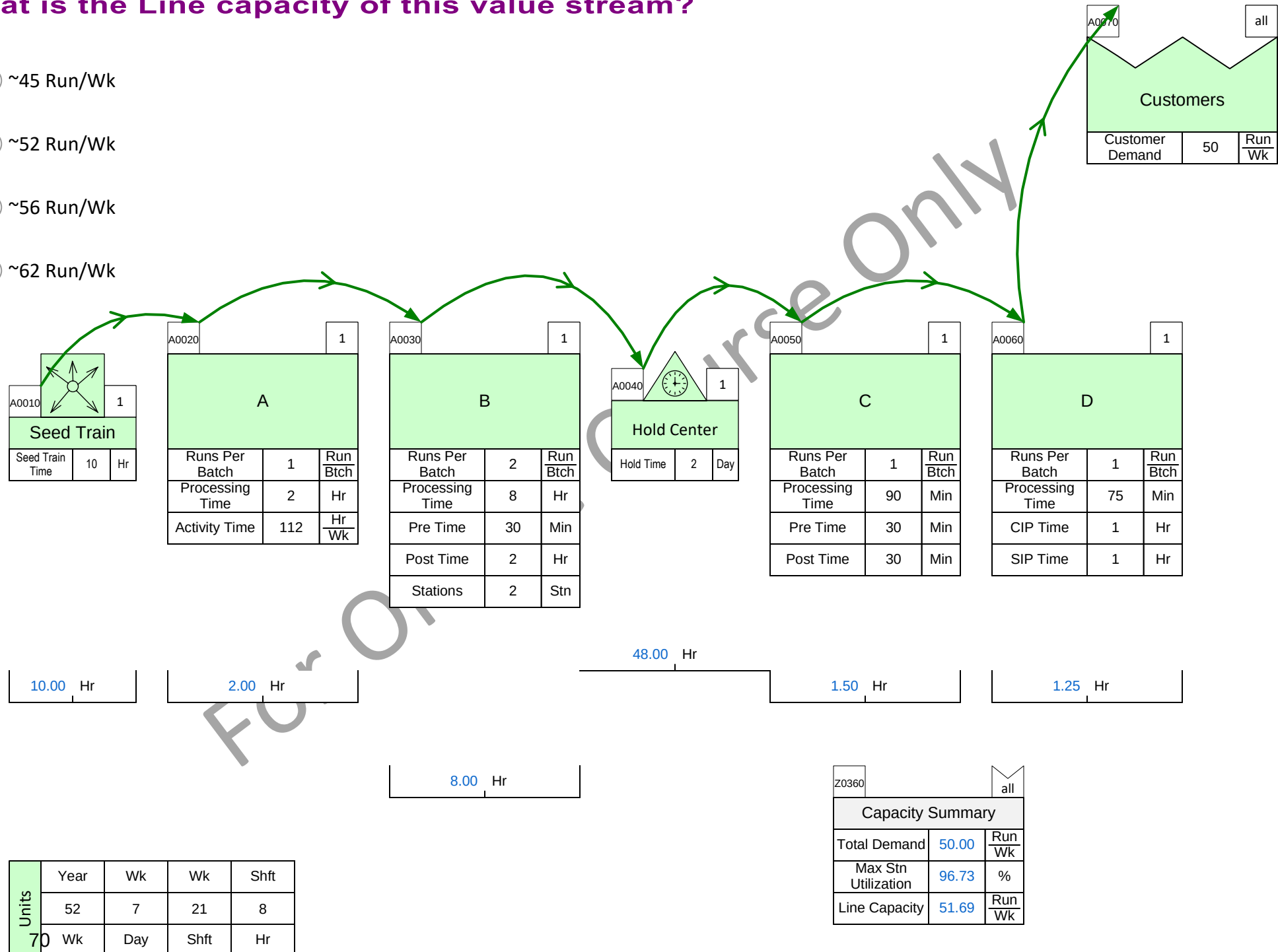
What is the Line capacity of this value stream?

☐ ~45 Run/Wk

☐ ~52 Run/Wk

☐ ~56 Run/Wk

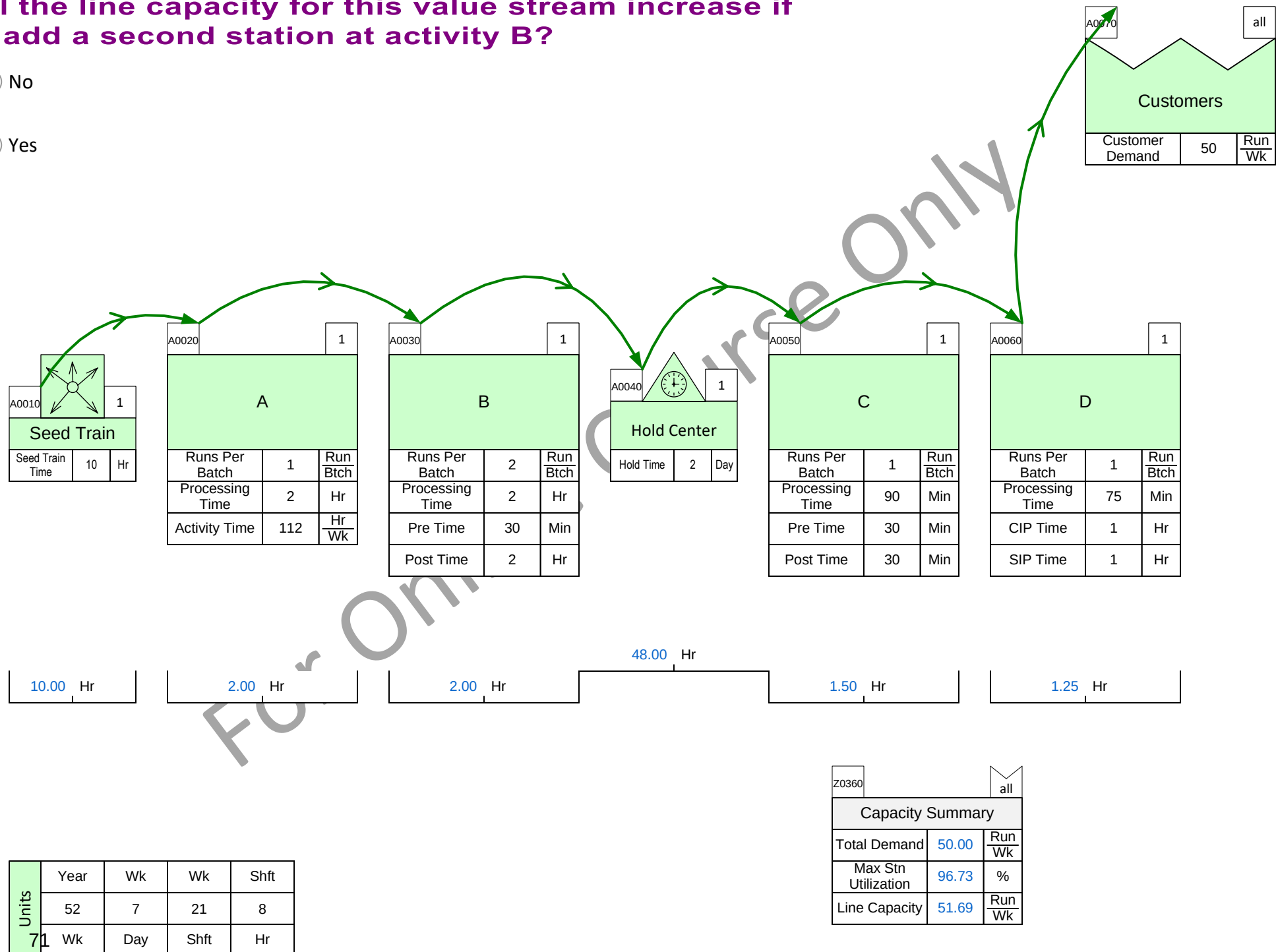
☐ ~62 Run/Wk



Will the line capacity for this value stream increase if we add a second station at activity B?

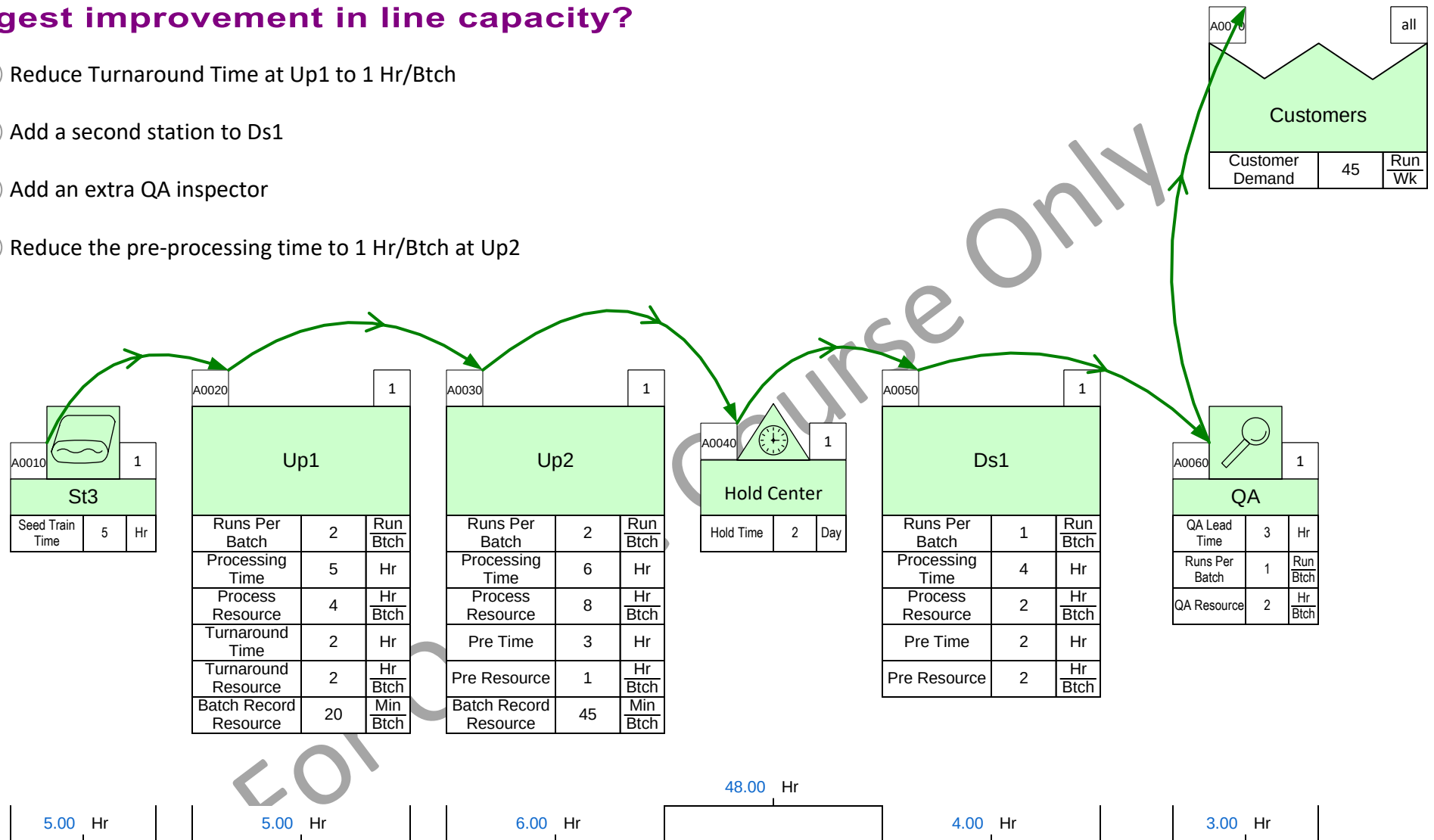
☐ No

☐ Yes



Which one of the following changes will show the biggest improvement in line capacity?

- ☐ Reduce Turnaround Time at Up1 to 1 Hr/Btch
- ☐ Add a second station to Ds1
- ☐ Add an extra QA inspector
- ☐ Reduce the pre-processing time to 1 Hr/Btch at Up2

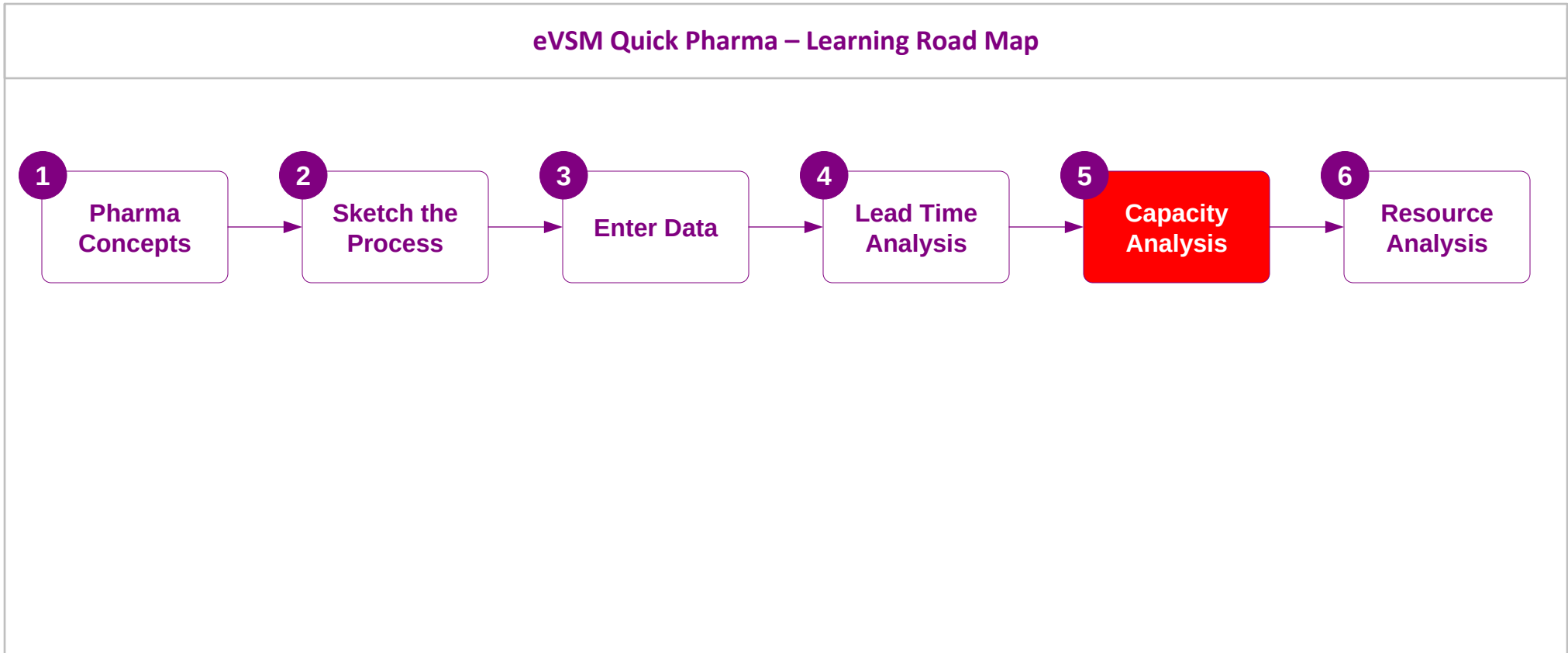


Units	Year	Wk	Wk	Shft
	52	7	21	8
72	Wk	Day	Shft	Hr

Capacity Summary		
Total Demand	45.00	Run Wk
Max Stn Utilization	160.71	%
Line Capacity	28.00	Run Wk

- You learned:**
- How to perform a capacity analysis on a map
 - How the Utilization chart can visualize capacity
 - How to show the line capacity
 - How to explore impact on capacity of improvement ideas

Road Map:

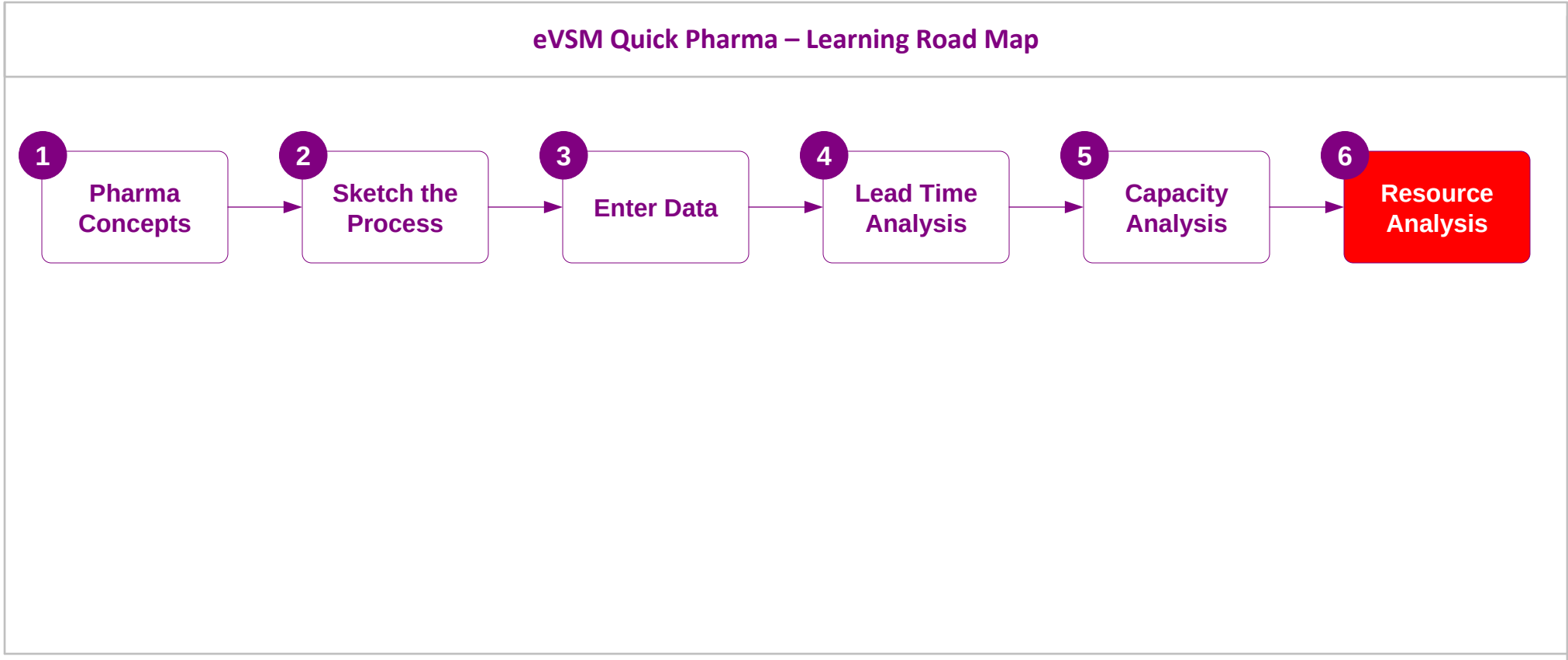


What's next:

You will learn how to perform a resource analysis on a map in eVSM.

Resource Analysis

Given your customer demand and operation parameters, eVSM can calculate resource required and the resource utilization for a part or the whole of the value stream. This lesson shows how.

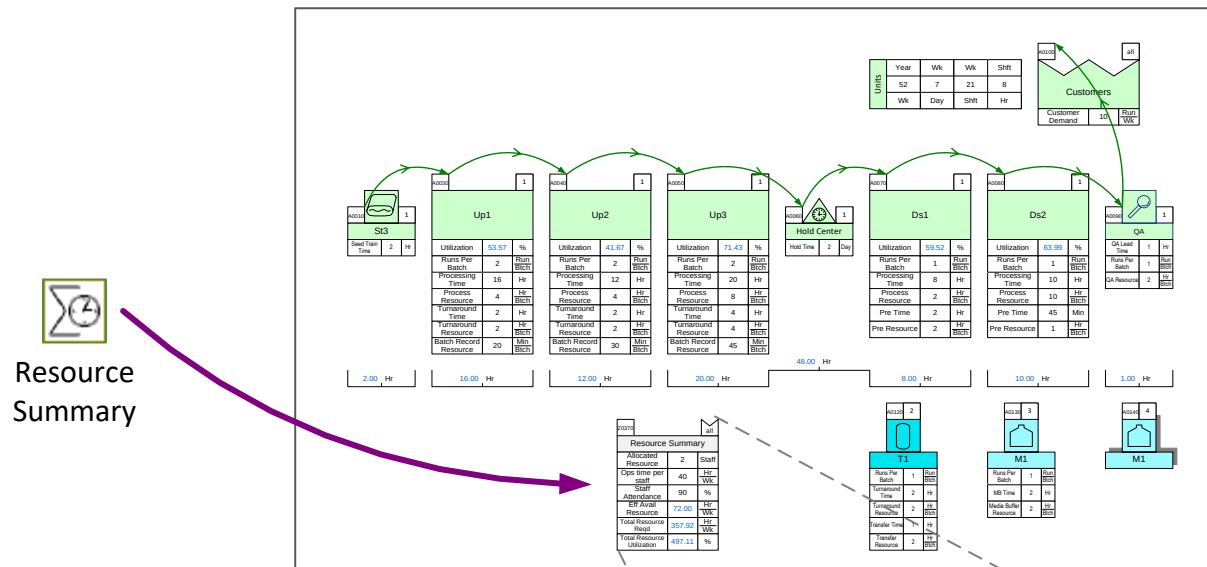


Resource analysis for the Pharma Process Flow

Total Resource Required and Total Resource Utilization

The total resource required for the value stream depends on the customer demand, the effort required to process each run, all the pre and post activities, and the supporting off-line activities. A quick way to get this total estimate is with the Resource Summary center.

Once you have the value stream model configured with the input data, simply drop the Resource Summary center on the page, enter the allocated staff data, and Solve.



Total Staff.

Staff availability after holidays, sick leave, etc. accounted for.

Z0020	Resource Summary		all
Allocated Resource	22	Staff	
Ops time per staff	40	Hr/Wk	
Staff Attendance	90	%	
Eff Avail Resource	792.00	Hr/Wk	
Total Resource Req'd	735.83	Hr/Wk	
Total Resource Utilization	92.91	%	

Blue values calculated automatically by eVSM.

Total Resource

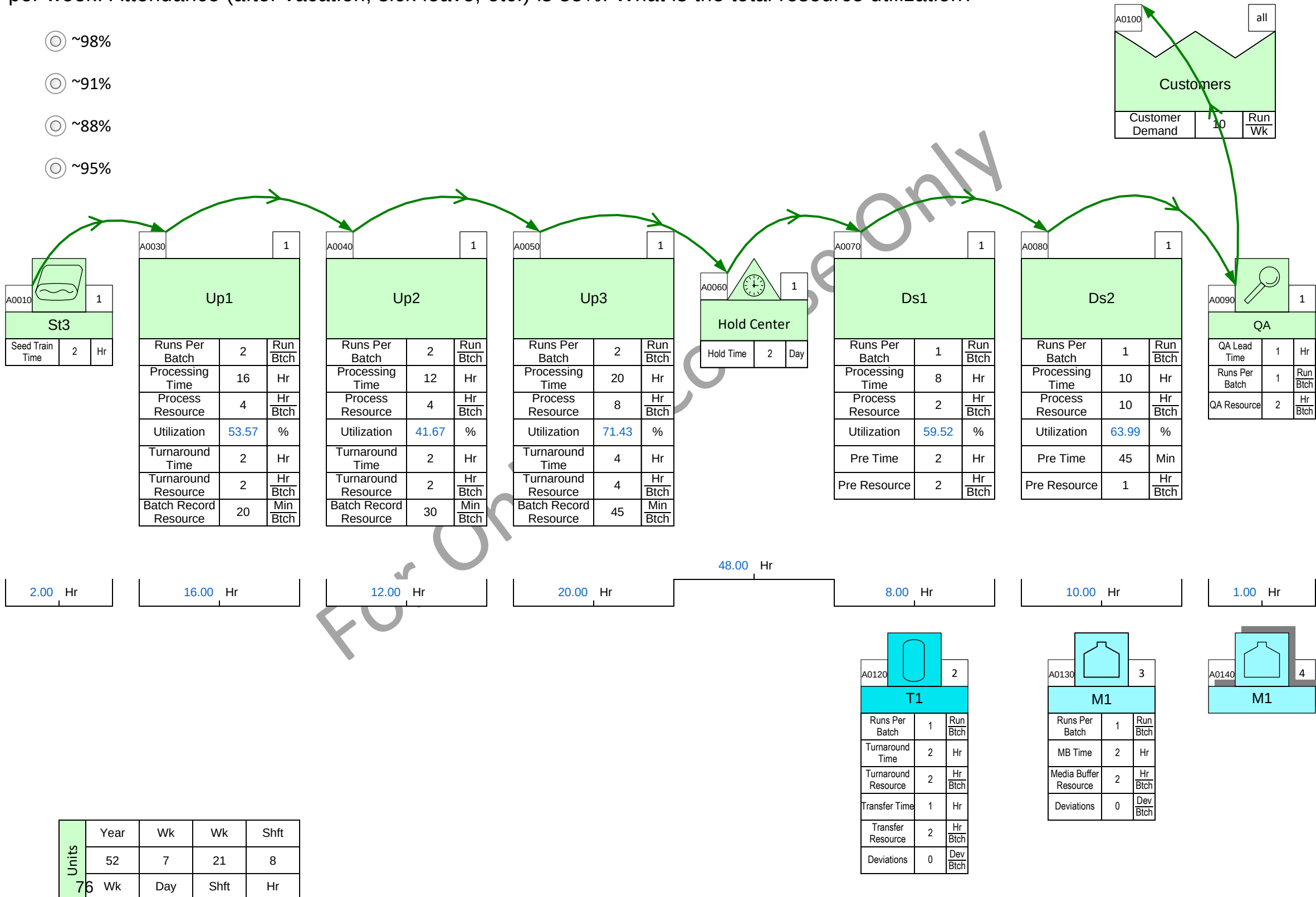
You have a team of 12 associates to cover all activities shown below. All associates work 40 hours per week. Attendance (after vacation, sick leave, etc.) is 85%. What is the total resource utilization?

☐ ~98%

☐ ~91%

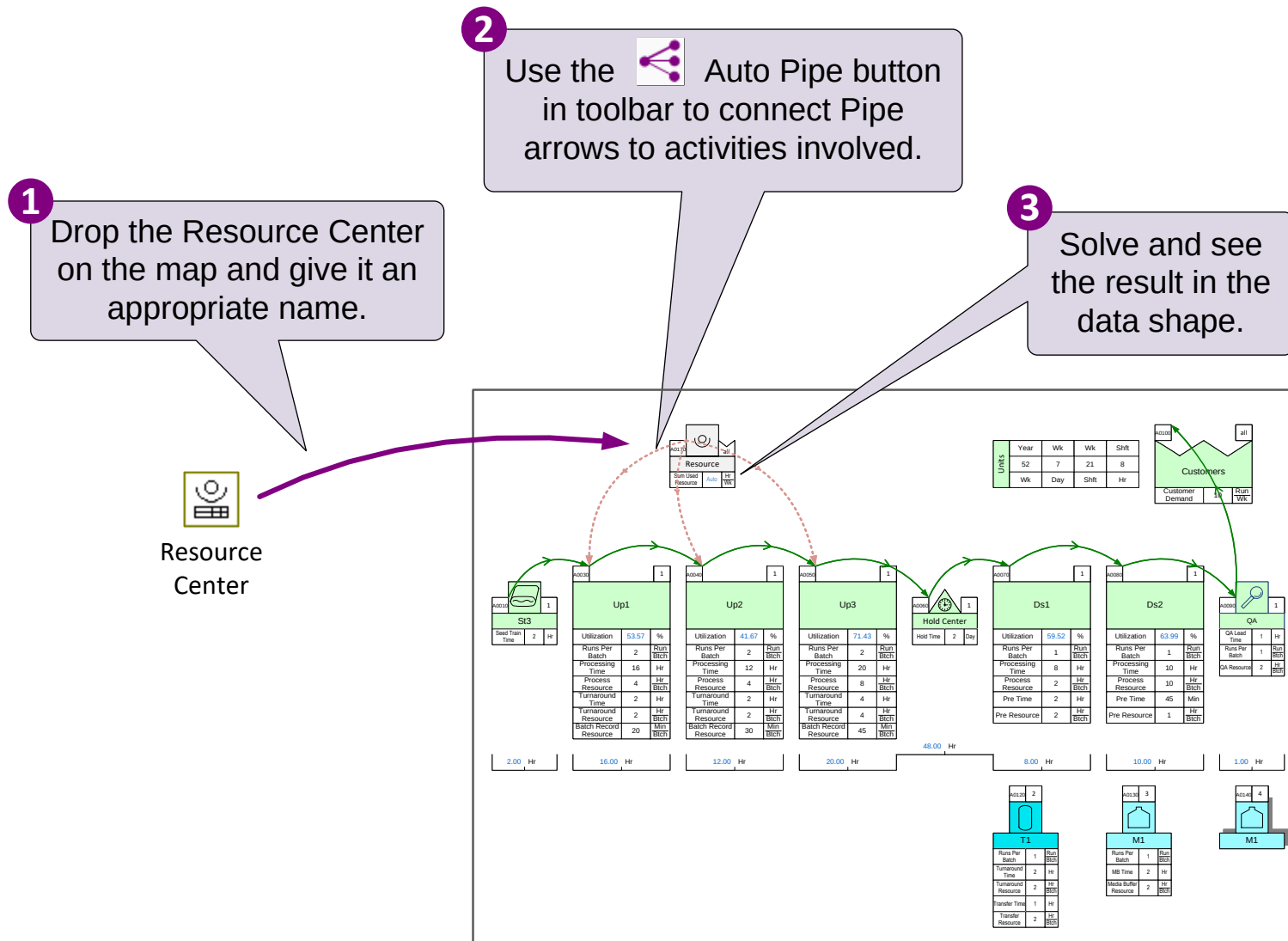
☐ ~88%

☐ ~95%



Resource Analysis for a Part of the Value Stream

When you need to calculate the resource usage for just a part of the value stream, you can use the Resource Center.

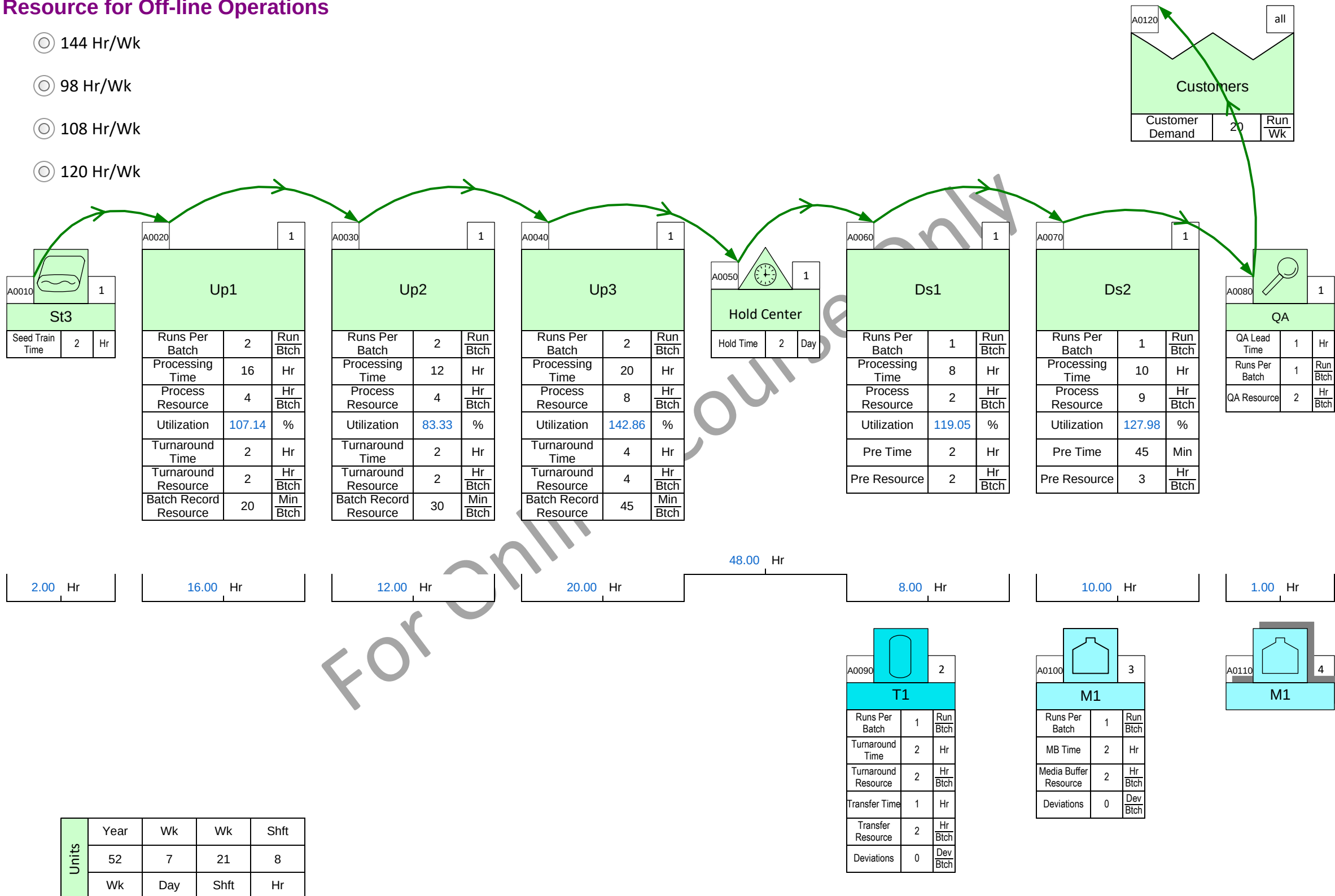


Resource for Off-line Operations

☐ 144 Hr/Wk

☐ 98 Hr/Wk

☐ 108 Hr/Wk

☐ 120 Hr/Wk


Resource Utilization for a Part of the Value Stream

To calculate the resource utilization for a part of the value stream, follow these steps.

- 1 Drop the Resource Center on the map and give it an appropriate name.



Resource Center

- 4 Use the  Auto Pipe button in toolbar to connect Pipe arrows to activities involved.

- 5 Solve and see the result in the Resource Center.



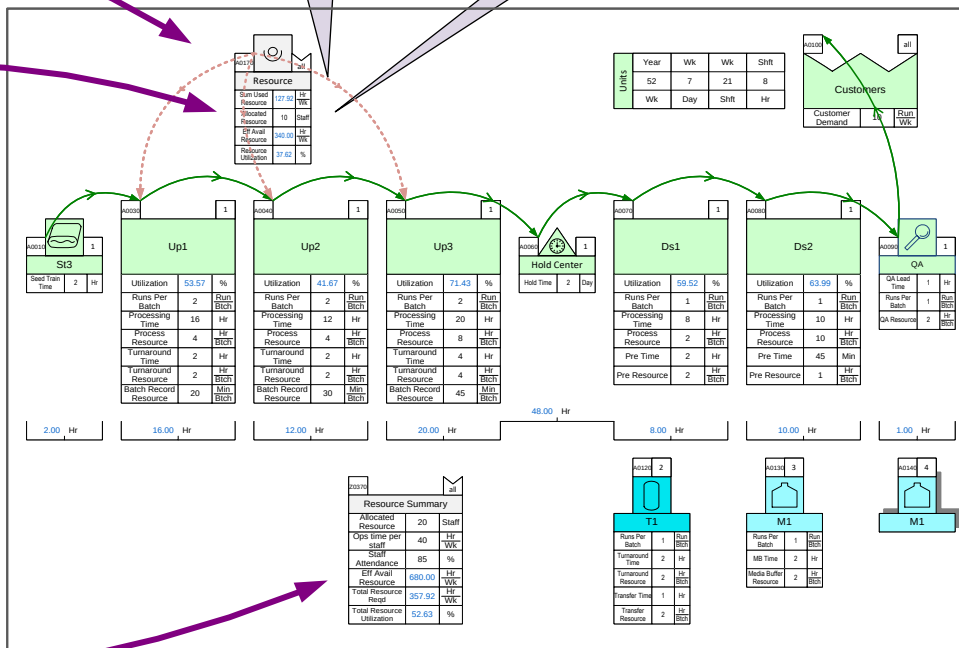
Resource Usage

- 2 Attach the Resource Usage add-on to the Resource Center.

- 3 Make sure there is a Resource Summary center on the map with the plant resource data.



Resource Summary



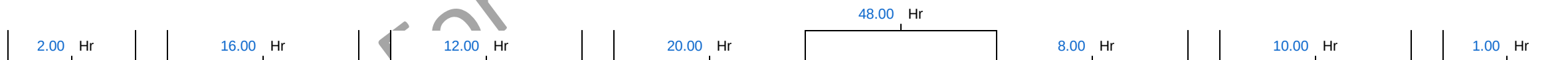
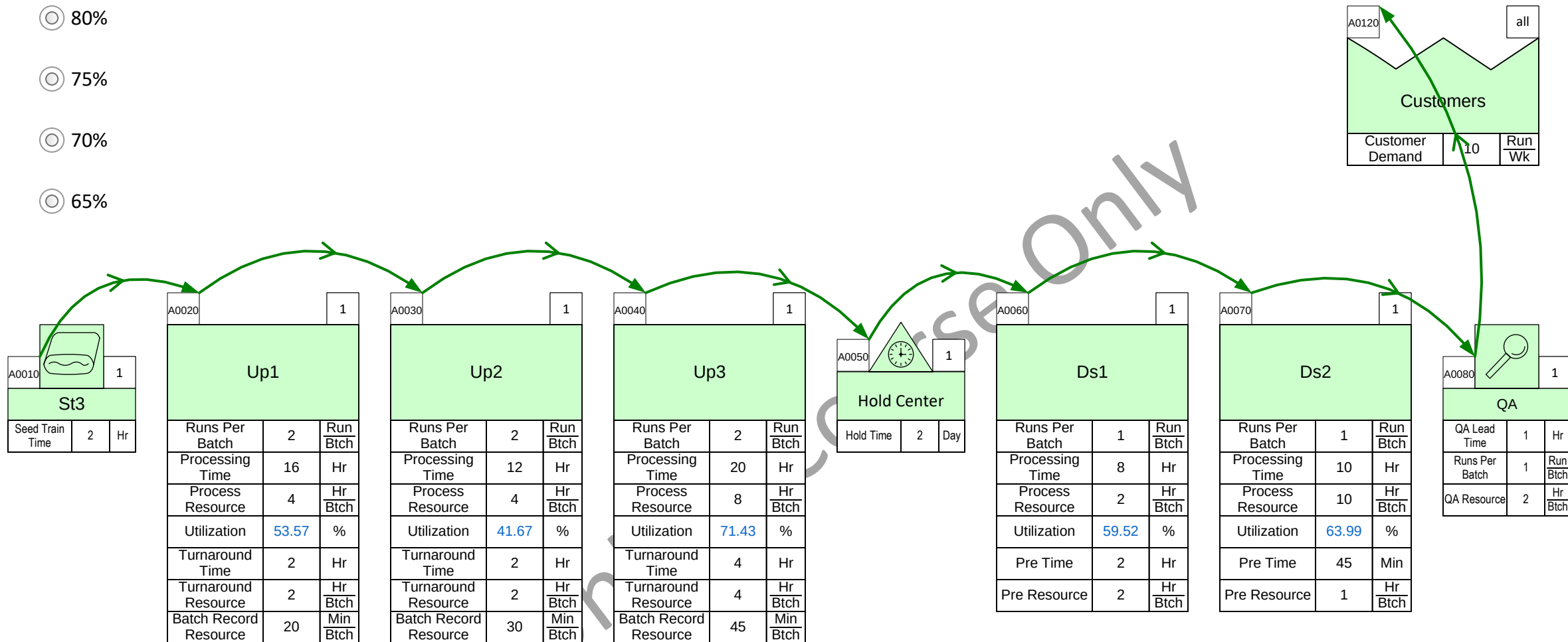
Resource for Up-stream In-line Operations

80%

75%

70%

65%



Units	Year	Wk	Wk	Shft
	52	7	21	8
	Wk	Day	Shft	Hr

Z0010			all
Resource Summary			
Allocated Resource	12	Staff	
Ops time per staff	40	Hr/Wk	
Staff Attendance	80	%	
Eff Avail Resource	384.00	Hr/Wk	
Total Resource Req'd	357.92	Hr/Wk	
Total Resource Utilization	93.21	%	

A0090		2
T1		
Runs Per Batch	1	Run Btch
Turnaround Time	2	Hr
Turnaround Resource	2	Hr Btch
Transfer Time	1	Hr
Transfer Resource	2	Hr Btch
Deviations	0	Dev Btch

A0100		3
M1		
Runs Per Batch	1	Run Btch
MB Time	2	Hr
Media Buffer Resource	2	Hr Btch
Deviations	0	Dev Btch

A0110		4
M1		

You learned:

- How resource requirements are modelled in eVSM
- How to calculate the total resource usage and utilization for the whole value stream
- How to calculate the resource usage and utilization for a part of the whole value stream

Recommended Next Steps:

1. Sketch your first value stream with the Quick Pharma stencil
2. Create a value stream model
3. Email any questions to support@evsm.com
4. Go through the eVSM Improvement Framework course - accessed from: <https://www.evsm.com/my-skills>

—Useful Links—

eVSM Toolbar Guide

evsm.com/toolbarguide

eVSM Productivity Guide

evsm.com/productivity

eVSM Blogs

evsm.com/blog

eVSM Support FAQ

evsm.com/support

Download the Latest Version

evsm.com/install